



All Life Bank Business Report

Pattern Discovery with Unsupervised Learning

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I. Problem Context:

All Life Bank is looking to strengthen its market position by focusing on its credit card portfolio for the upcoming financial year. Market research indicates significant opportunities for increased market penetration through personalized campaigns aimed at acquiring new customers and upselling to existing ones. Simultaneously, the bank faces challenges regarding customer perception of its support services, prompting the Operations team to seek an upgrade to the service delivery model to improve query resolution times.

To address these challenges, the bank has tasked the Data Science team with performing a customer segmentation analysis.

II. Objective:

Utilize clustering algorithms to categorize the existing customer base into distinct segments based on their unique spending behaviors and interaction histories with the bank. Provide recommendations to the bank on how to better market to and service these customers.

The goal is to translate these data-driven segments into actionable insights. These insights will enable the Marketing team to design targeted promotional campaigns and allow the Operations team to implement tailored service delivery strategies, ultimately improving customer satisfaction and engagement.

III. Problem Definition:

- Focus on the credit card customer base is not enough and All Life bank wants to focus this in next financial year.
- Penetration in the market is not sufficient and can be improved. Marketing team wants to run personalized campaigns to target new customers as well as upsell to existing customers.
- Customers perceive the support services of the bank poorly. This needs to be improved. Operations team wants to upgrade the service delivery model, to ensure that customer queries are resolved faster.

The goal is to translate these data-driven segments into actionable insights. These insights will enable the Marketing team to design targeted promotional campaigns and allow the Operations team to implement tailored service delivery strategies, ultimately improving customer satisfaction and engagement.

IV. Exploratory Data Analysis

IV.1. Data Description:

The data provided is of various customers of a bank and their financial attributes like credit limit, the total number of credit cards the customer has, and different channels through which customers have contacted the bank for any queries (including visiting the bank, online, and through a call center).

Data Dictionary:

- SI_No: Primary key of the records
- Customer Key: Customer identification number
- Average Credit Limit: Average credit limit of each customer for all credit cards
- Total credit cards: Total number of credit cards possessed by the customer
- Total visits bank: Total number of visits that the customer made (yearly) personally to the bank
- Total visits online: Total number of visits or online logins made by the customer (yearly)
- Total calls made: Total number of calls made by the customer to the bank or its customer service department (yearly)

IV.2. Data Quality Checks:

IV.2.1. Shape of Dataset

There are 660 rows and 7 columns

IV.2.2. Unique Values:

All fields except customer key are numerical in nature and hence there is no need to check uniqueness for those columns. For customer key, we will see if any non-unique values exist.

	SI_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
4	5	47437	100000	6	0	12	3
48	49	37252	6000	4	0	2	8
104	105	97935	17000	2	1	2	10
332	333	47437	17000	7	3	1	0
391	392	96929	13000	4	5	0	0
398	399	96929	67000	6	2	2	2
411	412	50706	44000	4	5	0	2

	SI_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
432	433	37252	59000	6	2	1	2
541	542	50706	60000	7	5	2	2
632	633	97935	187000	7	1	7	0

From the above, we can see the data is different for the unique key. The 'Customer Key' is a unique ID given to each customer in the database. The duplicate values might correspond to customer profile changes, and as such, there is no need to delete these records as these are actual occurrences at some point in the time.

IV.2.3. Missing Values:

There are 660 rows and 7 columns in the dataset. There are no missing values in data

IV.2.4. Duplicate Values:

There are no duplicate values in data

IV.2.5. Five Number Summary:

There are **5-Number Summary** for the data -A set of five descriptive statistics that summarize a distribution: **Minimum, Q1 (25th percentile), Median (50%), Q3 (75th percentile), and Maximum.**

	count	mean	std	min	25%	50%	75%	max
SI_No	660.0	330.500000	190.669872	1.0	165.75	330.5	495.25	660.0
Customer Key	660.0	55141.44393	25627.77220	11265.0	33825.25	53874.5	77202.50	99843.0
Avg_Credit_Limit	660.0	34574.24242	37625.48780	3000.0	10000.0	18000.0	48000.0	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.00	5.0	6.00	10.0

	count	mean	std	min	25%	50%	75%	max
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.00	2.0	4.00	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.00	2.0	4.00	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.00	3.0	5.00	10.0

Inference from Statistical Data:

- **Data Types:** All columns are of integer type (int64), which is appropriate for numerical analysis. This ensures there are no issues with mixed data types or incorrect interpretations.
- **Missing Values:** As previously observed, there are no missing values in the dataset, indicating a clean dataset ready for analysis.
- **Identifier Columns:** SI_No and Customer Key are unique identifiers for each record and customer, respectively. These columns are not suitable for direct use in clustering algorithms as they do not represent behavioral or financial attributes.
- **Average Credit Limit (Avg_Credit_Limit):**
 - The credit limits range from 3,000 to 200,000.
 - The mean (34,574) is significantly higher than the median (18,000), and the 75th percentile is 48,000. This indicates a right-skewed distribution, meaning a larger number of customers have lower credit limits, while a smaller group has considerably higher limits.
- **Total Credit Cards (Total_Credit_Cards):**
 - Customers possess between 1 and 10 credit cards, with an average of approximately 4.7 cards.
 - The median is 5, suggesting a fairly even distribution around the mean.
- **Total Visits Bank (Total_visits_bank):**
 - Bank visits range from 0 to 5 times a year.
 - The average is around 2.4 visits, with the median also at 2. Most customers visit the bank a few times annually.
- **Total Visits Online (Total_visits_online):**
 - Online visits range from 0 to 15 times a year.

- The mean is about 2.6 visits, with the median at 2. This shows that while most customers engage online a few times, some customers are very active online, making up to 15 visits.
- **Total Calls Made (Total_calls_made):**
 - Customers make between 0 and 10 calls annually.
 - The average is approximately 3.58 calls, with the median at 3. This indicates a moderate level of customer interaction through calls.

IV.2.6. Data Types

All are numerical integer data. Memory usage looks fine.

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	SI_No	660 non-null	int64
1	Customer Key	660 non-null	int64
2	Avg_Credit_Limit	660 non-null	int64
3	Total_Credit_Cards	660 non-null	int64
4	Total_visits_bank	660 non-null	int64
5	Total_visits_online	660 non-null	int64
6	Total_calls_made	660 non-null	int64

dtypes: int64(7)

memory usage: 36.2 KB

IV.3. Univariate Analysis:

IV.3.1. Univariate Analysis For Avg_Credit_Limit:

IV.3.1.1. Descriptive statistics for Avg_Credit_Limit:

Descriptive statistics for Avg_Credit_Limit:	
count	660.000000
mean	34574.242424
std	37625.487804
min	3000.000000
25%	10000.000000
50%	18000.000000
75%	48000.000000
max	200000.000000

```
Name: Avg_Credit_Limit, dtype: float64
```

IV.3.1.2. Visual Representation for Avg_Credit_Limit:

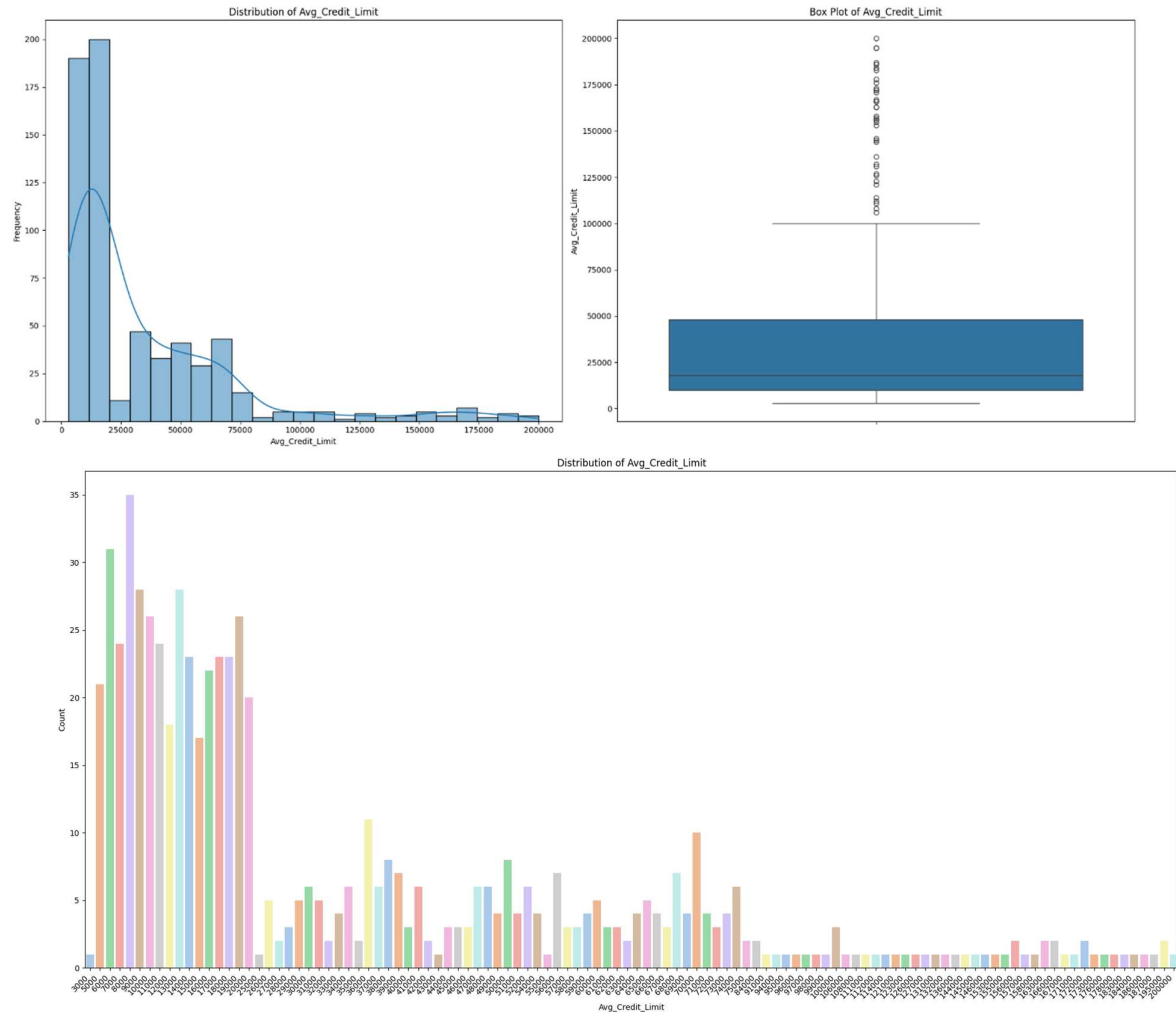


Figure 1 : Visual - Avg_Credit_Limit

IV.3.2. Univariate Analysis For Total_Credit_Cards

IV.3.2.1. Descriptive statistics for Total_credit_Cards

Descriptive statistics for Total_Credit_Cards:	
count	660.000000
mean	4.706061
std	2.167835
min	1.000000
25%	3.000000
50%	5.000000
75%	6.000000
max	10.000000

```
Name: Total_Credit_Cards, dtype: float64
```

IV.3.2.2. Visual Representation for Total_credit_Cards

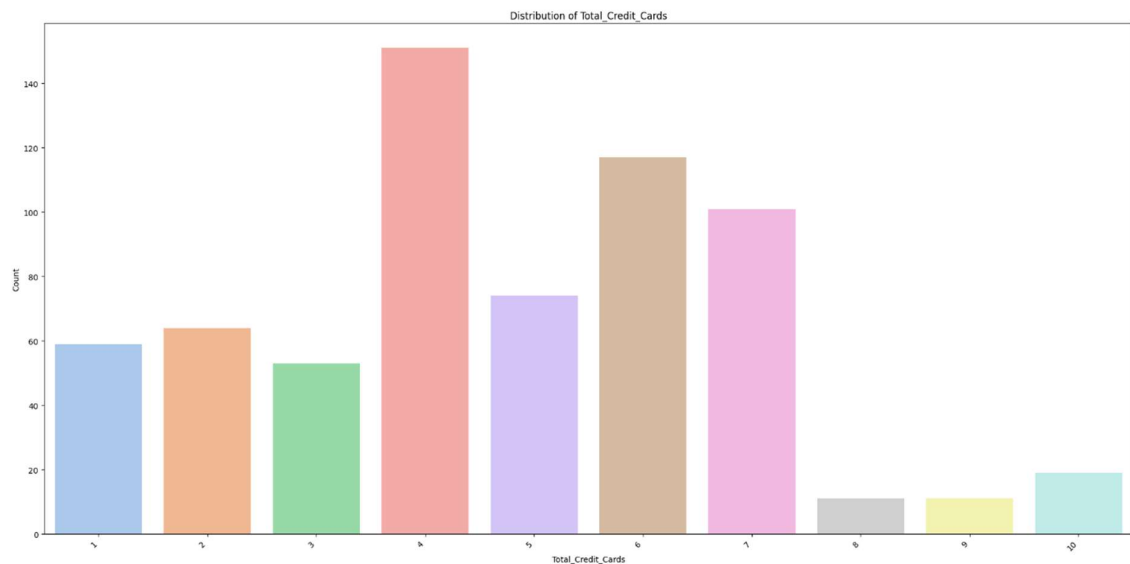
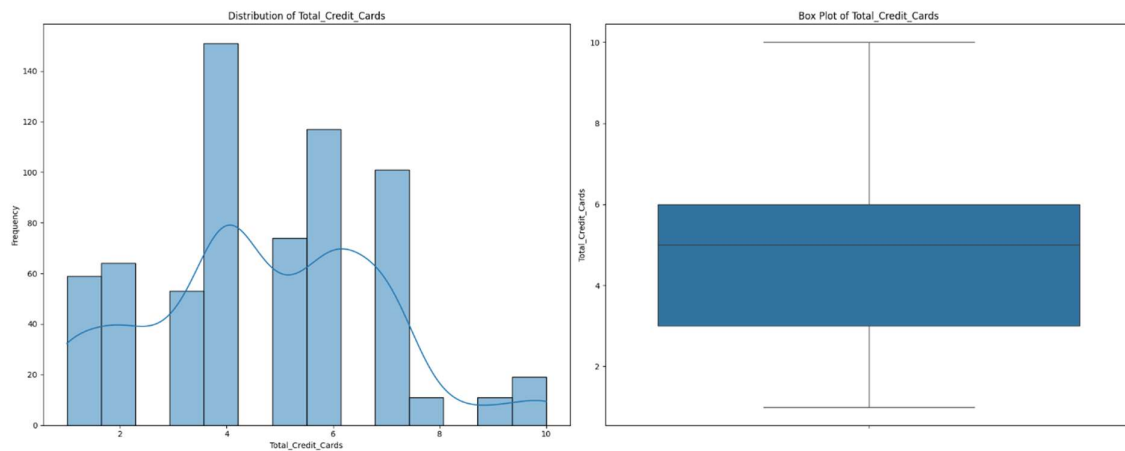


Figure 2 Univariate - Total Credit Cards

IV.3.3. Univariate Analysis For Total_Visits_Bank

IV.3.3.1. Descriptive statistics for Total_Visits_Bank

Descriptive statistics for Total_visits_bank:	
count	660.000000
mean	2.403030
std	1.631813
min	0.000000
25%	1.000000
50%	2.000000
75%	4.000000

max	5.000000
Name: Total_visits_bank, dtype: float64	

IV.3.3.2. Visual Representation for Total_Visits_Bank

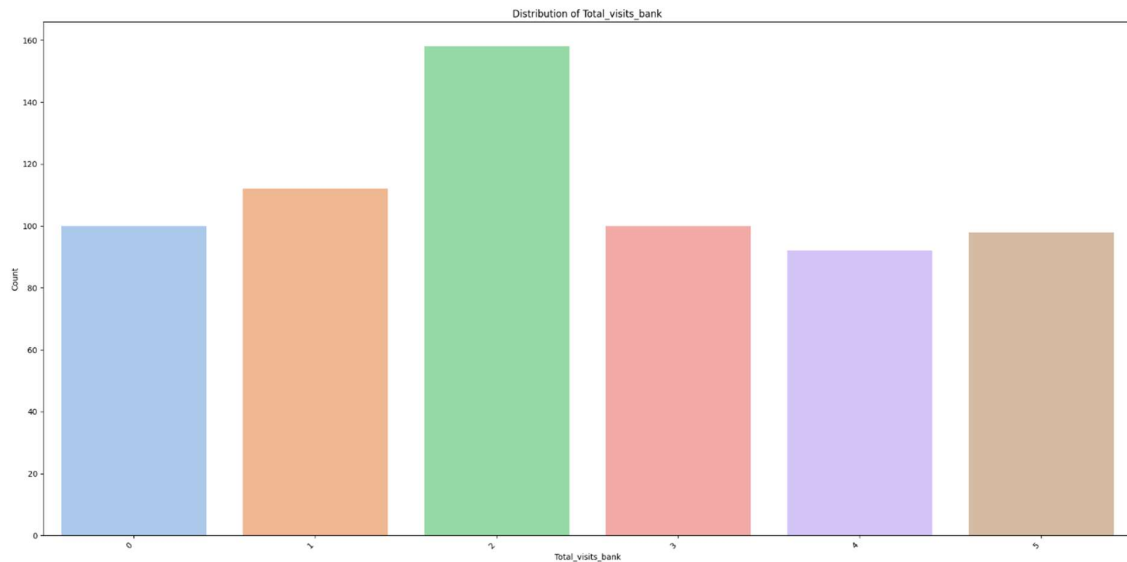
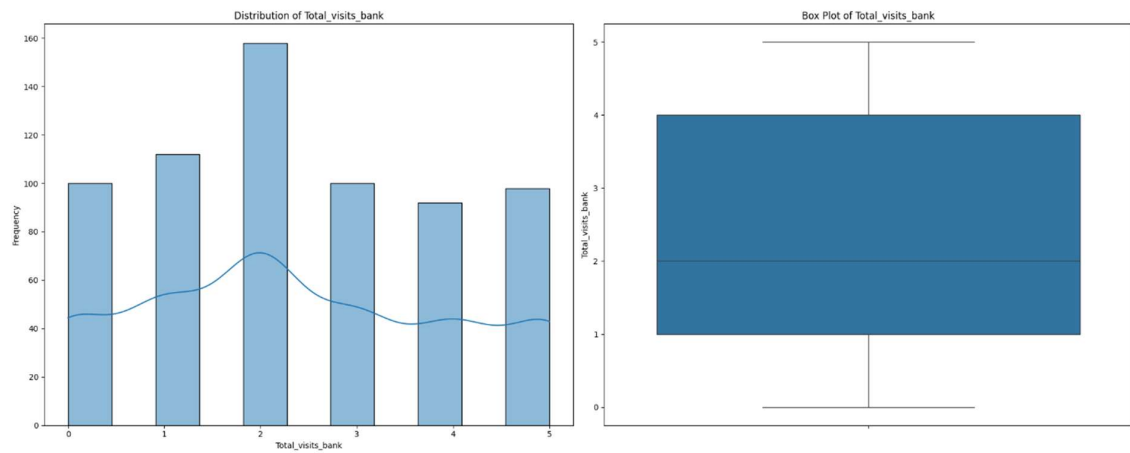


Figure 3 Univariate - Total Visits Bank

IV.3.4. Univariate Analysis For Total_Visits_Online

IV.3.4.1. Descriptive statistics for Total_Visits_Online

Descriptive statistics for Total_visits_online:	
---	--

count	660.000000
mean	2.606061
std	2.935724
min	0.000000
25%	1.000000
50%	2.000000
75%	4.000000

max	15.000000
Name: Total_visits_online, dtype: float64	

IV.3.4.2. Visual Representation for Total_Visits_Online

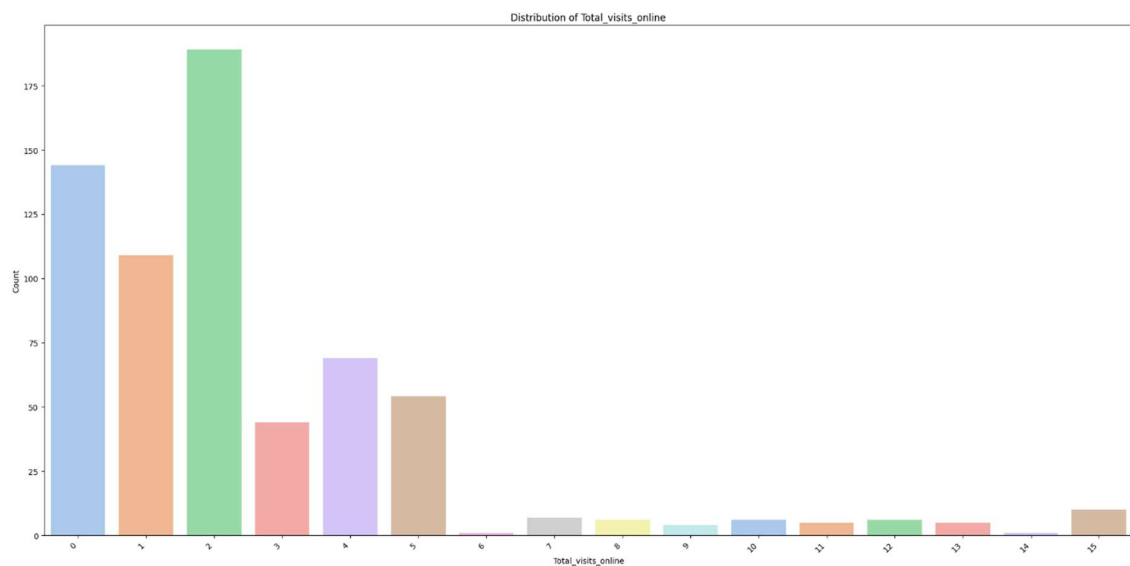
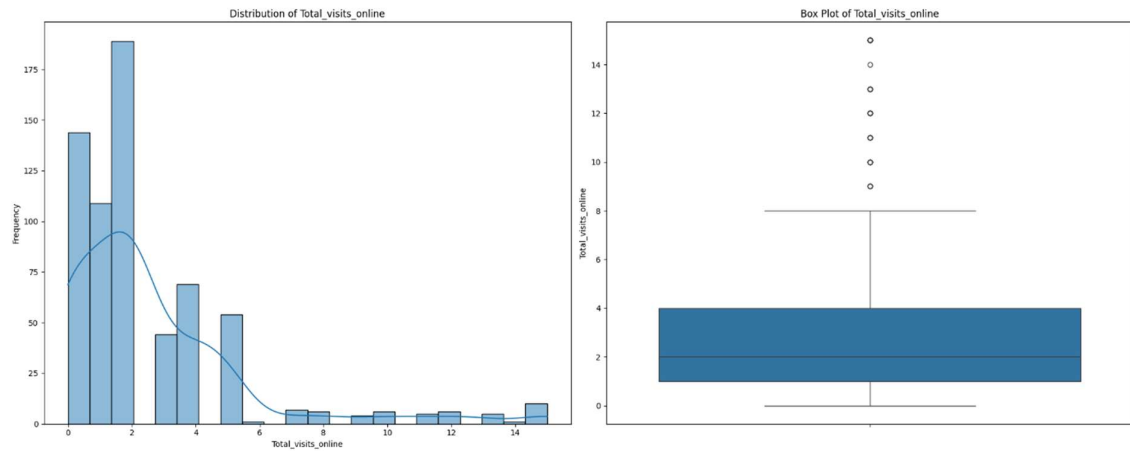


Figure 4 Univariate - Total Visits Online

IV.3.5. Univariate Analysis For Total_Calls_Made

IV.3.5.1. Descriptive statistics for Total_Calls_Made

Descriptive statistics for Total_calls_made:	
count	660.000000
mean	3.583333
std	2.865317
min	0.000000
25%	1.000000
50%	3.000000

75%	5.000000
max	10.000000
Name: Total_calls_made, dtype: float64	

IV.3.5.2. Visual Representation for Total_Calls_Made

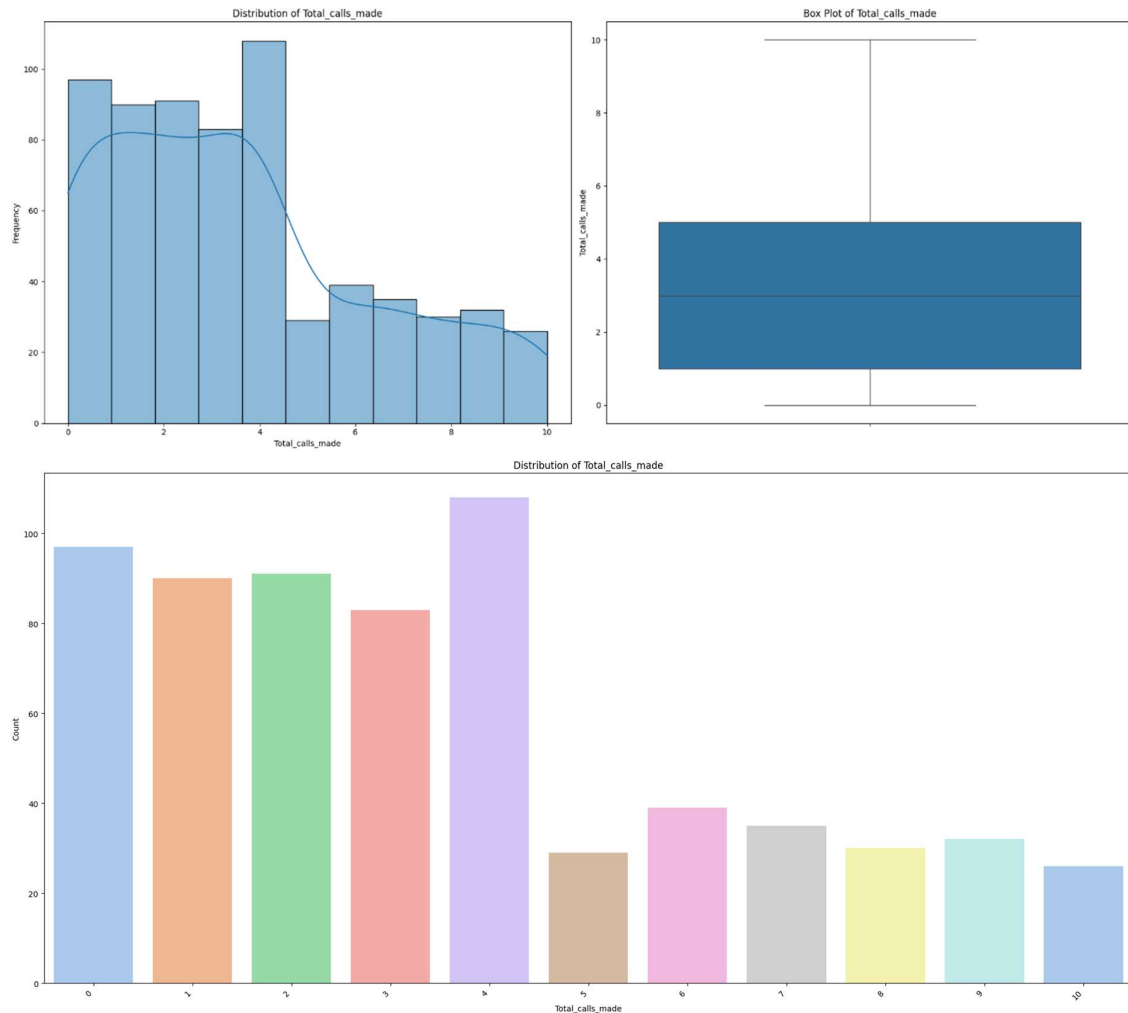


Figure 5 Univariate - Total Calls Made

IV.4. Inferences from Univariate Analysis

1. Average Credit Limit (Avg_Credit_Limit):

- **Distribution:** The histogram and descriptive statistics show a right-skewed distribution. A large number of customers have low to moderate credit limits (below 50,000), while a smaller proportion have very high credit limits (up to 200,000).
- **Central Tendency & Spread:** The mean (34,574) is considerably higher than the median (18,000), confirming the right-skewness. The standard deviation (37,625) is also large, indicating significant variability in credit limits.

- Outliers: The box plot clearly shows many outliers on the higher end, indicating a segment of customers with exceptionally high credit limits compared to the majority.
 - Count Plot: For this continuous variable, the count plot reinforces the right-skewness, showing fewer observations as the credit limit increases.
2. Total Credit Cards (Total_Credit_Cards):
- Distribution: The distribution appears somewhat balanced, with a slight peak around 3-7 credit cards. Most customers hold between 1 and 7 cards.
 - Central Tendency & Spread: The mean (4.7) and median (5) are very close, suggesting a relatively symmetric distribution for the number of credit cards. The range is from 1 to 10.
 - Outliers: The box plot shows no significant outliers, indicating that the number of credit cards held by customers is mostly within a typical range.
 - Count Plot: This plot effectively visualizes the frequency of each number of credit cards, showing common patterns like many customers having 3, 5, or 6 cards.
3. Total Visits Bank (Total_visits_bank):
- Distribution: The distribution is discrete and shows that a significant portion of customers visit the bank 0, 1, or 2 times. There's a gradual decrease in frequency as the number of visits increases.
 - Central Tendency & Spread: The mean is 2.4 visits, with the median at 2. The maximum visits are 5, indicating that very few customers visit the bank frequently.
 - Outliers: The box plot shows no significant outliers, suggesting that the observed visit counts are within expected bounds for the dataset.
 - Count Plot: This is highly effective, showing clear bars for each visit count, with 1 visit being the most frequent, followed by 0 and 2 visits.
4. Total Visits Online (Total_visits_online):
- Distribution: Similar to bank visits, online visits are discrete. There's a tendency for customers to have 0, 1, or 2 online visits, but there's a long tail extending up to 15 visits.
 - Central Tendency & Spread: The mean is 2.6 visits, with the median at 2. The maximum is 15, which is quite high compared to the average.
 - Outliers: The box plot indicates several outliers on the higher end, suggesting some customers are highly active online.
 - Count Plot: This plot highlights that while many customers have low online activity, a smaller group engages very frequently online, which aligns with the outliers seen in the box plot.
5. Total Calls Made (Total_calls_made):
- Distribution: The distribution for calls made is also discrete, showing peaks at lower call counts (0-2 calls), and then a decreasing frequency as the number of calls increases.
 - Central Tendency & Spread: The mean is 3.58 calls, with the median at 3. The maximum calls are 10.
 - Outliers: The box plot indicates no significant outliers, suggesting a consistent pattern of customer calls within the observed range.
 - Count Plot: This plot provides a clear view of the frequency of customers making different numbers of calls, with 0, 1, and 2 calls being very common.

IV.5. BiVariate Analysis

IV.5.1. Correlation Heatmap of Numerical Columns

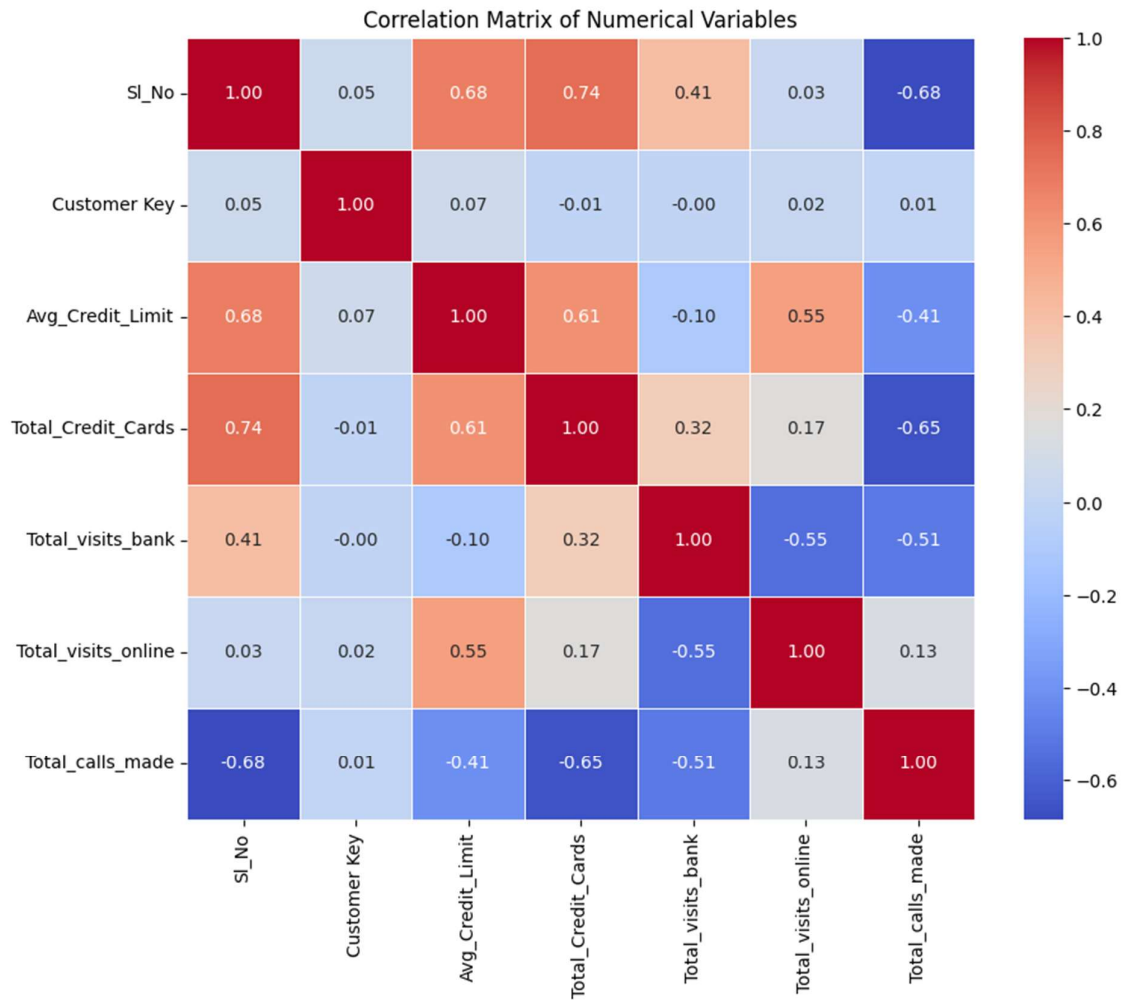


Figure 6 Correlation Map- Bivariate

Table 1 Correlation Matrix

	SI_No	Custo mer Key	Avg_Credit_ Limit	Total_Credit_ Cards	Total_visits _bank	Total_visits_ online	Total_calls_ made
SI_No	1.000 000	0.0528 86	0.677962	0.739329	0.406438	0.033916	-0.684125
Customer Key	0.052 886	1.0000 00	0.068604	-0.010281	-0.000560	0.022506	0.005968

	SI_No	Custo mer Key	Avg_Credit_ Limit	Total_Credit_ Cards	Total_visits_ _bank	Total_visits_ online	Total_calls_ made
Avg_Credit_L imit	0.677 962	0.0686 04	1.000000	0.608860	-0.100312	0.551385	-0.414352
Total_Credit_ Cards	0.739 329	- 0.0102 81	0.608860	1.000000	0.315796	0.167758	-0.651251
Total_visits_ bank	0.406 438	- 0.0005 60	-0.100312	0.315796	1.000000	-0.551861	-0.506016
Total_visits_ online	0.033 916	0.0225 06	0.551385	0.167758	-0.551861	1.000000	0.127299
Total_calls_ made	- 0.684 125	0.0059 68	-0.414352	-0.651251	-0.506016	0.127299	1.000000

IV.5.2. Inferences from Correlation Matrix

1. **Strong Positive Correlation:** Avg_Credit_Limit and Total_Credit_Cards (0.61)
 - This indicates a moderately strong positive relationship: as a customer's average credit limit increases, they tend to possess more credit cards, and vice versa. This is an expected and logical relationship, as banks often offer higher credit limits to customers with multiple cards or a good credit history.
2. **Moderate Positive Correlation:** Avg_Credit_Limit and Total_visits_online (0.55)
 - Customers with higher average credit limits tend to have more online visits. This could imply that customers with higher credit limits are more digitally active in managing their accounts or exploring banking services online.
3. **Strong Negative Correlation:** Total_calls_made and Total_Credit_Cards (-0.65)
 - There is a strong inverse relationship between the number of calls made and the total credit cards a customer has. Customers with more credit cards tend to make fewer calls to the bank. This might suggest that customers with more cards are more self-sufficient in managing their finances, or they might be more experienced and require less assistance.
4. **Moderate Negative Correlation:** Total_visits_bank and Total_visits_online (-0.55)
 - This shows a moderate inverse relationship: customers who visit the bank more frequently tend to visit online less, and vice versa. This suggests a behavioral segmentation where customers prefer either physical banking or digital banking channels.

5. **Moderate Negative Correlation:** `Total_calls_made` and `Avg_Credit_Limit` (-0.41)

- Customers with higher average credit limits tend to make fewer calls to the bank. This aligns with the idea that financially more stable or experienced customers might require less direct assistance.

6. **Weak/Negligible Correlations:**

- `Customer Key` and `SI_No` have very low or negligible correlations with other variables, as expected, since they are identifiers and not behavioral or financial metrics.
- Most other correlations are weak (close to 0), indicating that those pairs of variables do not have a strong linear relationship. For example, `Total_visits_bank` and `Total_Credit_Cards` have a weak positive correlation (0.32), suggesting that having more cards doesn't strongly dictate bank visits.

IV.5.3. Pairplot for Bivariate Analysis

A pairplot visualize pairwise relationships between variables, as well as the distribution of each variable.

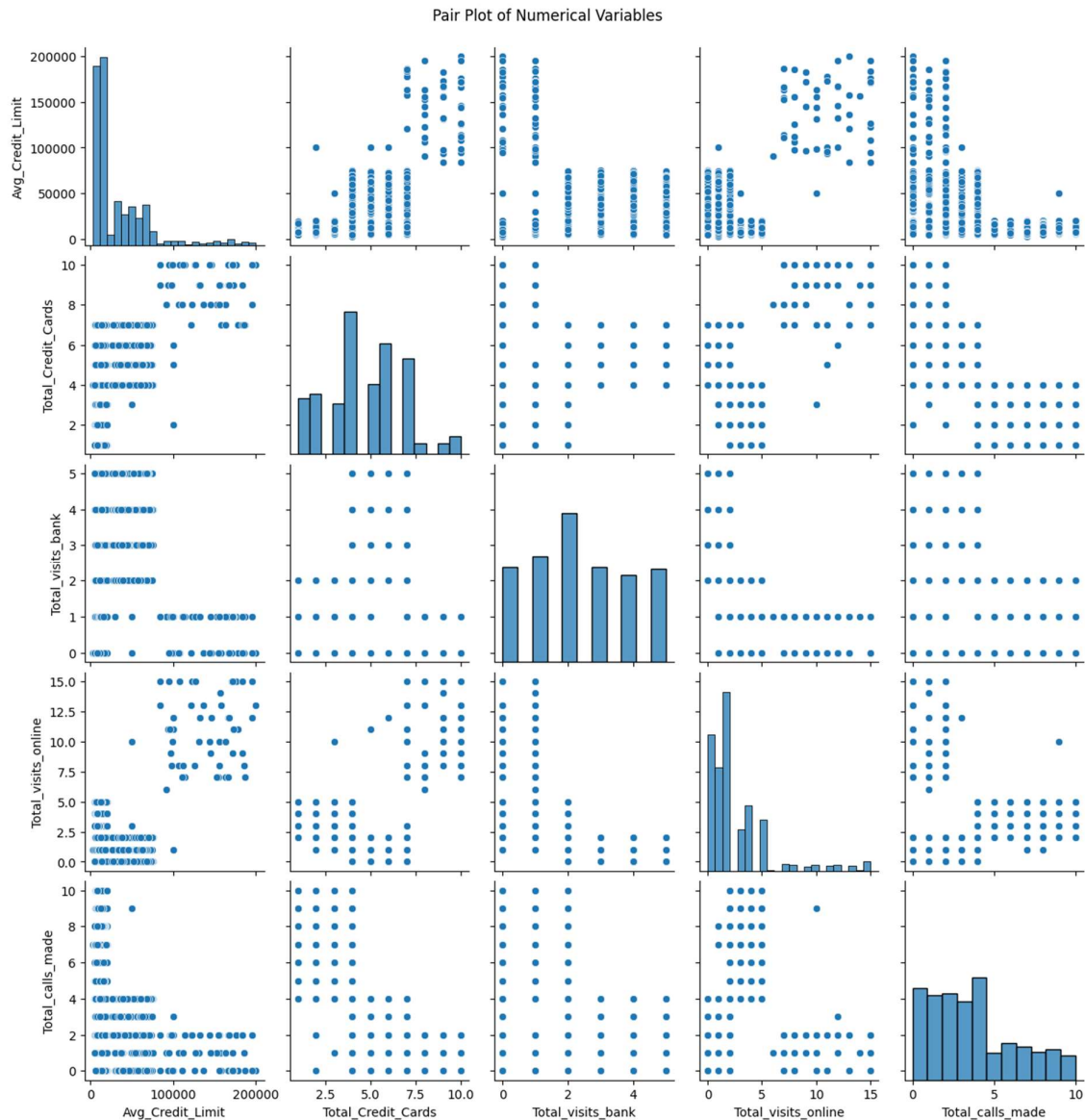


Figure 7 PairPlot

IV.5.4. Inferences from Pair Plot:

1. Diagonal Plots (Histograms/KDEs):

- The diagonal plots show the univariate distribution of each numerical variable. These confirm observations from the earlier univariate analysis:
 - **Avg_Credit_Limit** is right-skewed, indicating more customers with lower credit limits.
 - **Total_Credit_Cards** shows a somewhat normal distribution, centered around 4-6 cards.
 - **Total_visits_bank**, **Total_visits_online**, and **Total_calls_made** are discrete and show distributions concentrated at lower values, with some customers having higher activity (especially online visits).

2. Off-Diagonal Scatter Plots (Bivariate Relationships):

- **Avg_Credit_Limit** vs. **Total_Credit_Cards**: The scatter plot shows a clear positive linear relationship, reaffirming that customers with higher credit limits tend to have more credit cards. The points are not perfectly aligned, indicating some variability but a general trend.
- **Avg_Credit_Limit** vs. **Total_visits_online**: A positive, but less pronounced, linear relationship is visible. Customers with higher credit limits tend to visit online more, but there's a wider spread of data points.
- **Total_calls_made** vs. **Total_Credit_Cards**: This plot vividly displays the strong negative correlation. As the number of credit cards increases, the number of calls made generally decreases. This implies that customers with more cards might be more experienced and require less assistance.
- **Total_visits_bank** vs. **Total_visits_online**: A clear negative correlation is observed. As bank visits increase, online visits tend to decrease, and vice versa. This visually confirms the inverse relationship, suggesting customers tend to prefer one channel over the other.
- **Total_calls_made** vs. **Avg_Credit_Limit**: The negative correlation is evident, albeit with more scatter. Higher credit limits are associated with fewer calls.

3. General Observations:

- The pairplot helps visualize not just the presence, but also the strength and linearity of the correlations identified in the correlation matrix.
- Many relationships, while statistically correlated, show a fair amount of spread, indicating that while trends exist, they are not perfectly predictive.
- The plots for discrete variables (e.g., **Total_Credit_Cards**, **Total_visits_bank**) show distinct clusters or lines of points rather than continuous distributions, as expected.

IV.5.5. Joint Plot for Bivariate Analysis

create a joint plot for **Avg_Credit_Limit** and **Total_Credit_Cards** to see their joint distribution and individual distributions. We can use a scatter plot for the main bivariate relationship and histograms for the marginal distributions.

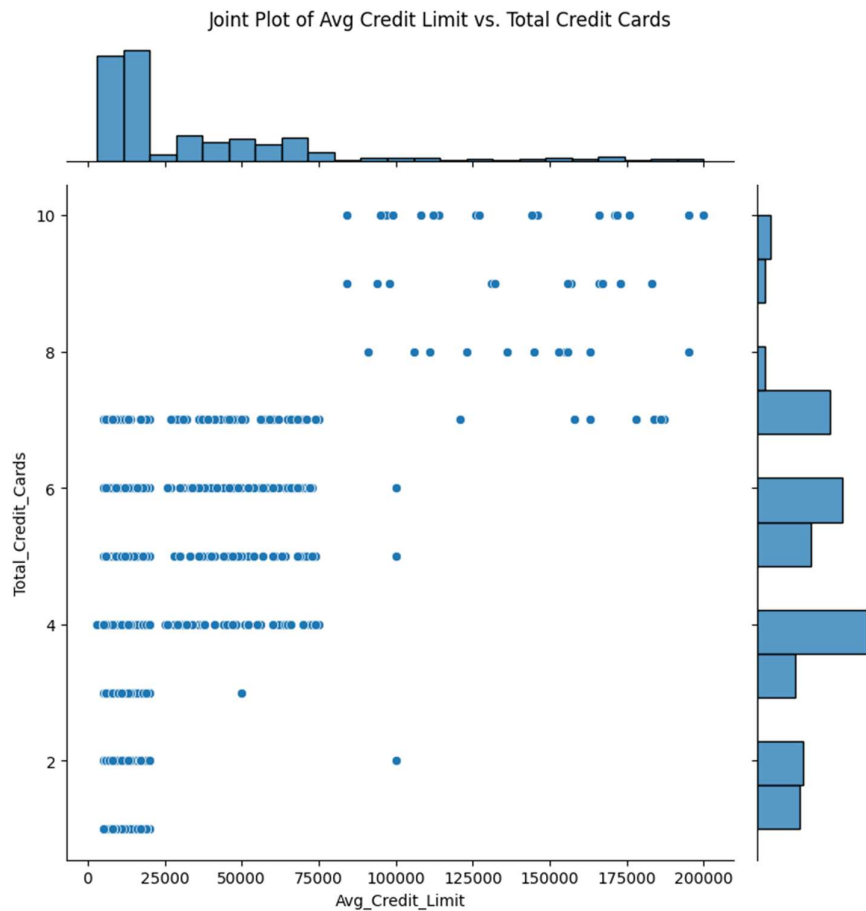


Figure 8 Joint Plot- Avg Credit Limit vs Total Credit Cards

We can also use a different kind for the joint plot, like kde for kernel density estimates or hex for hexbin plots, especially useful for visualizing density in large datasets or where overplotting is an issue. Let's try a hexbin plot for Total_visits_bank and Total_visits_online.

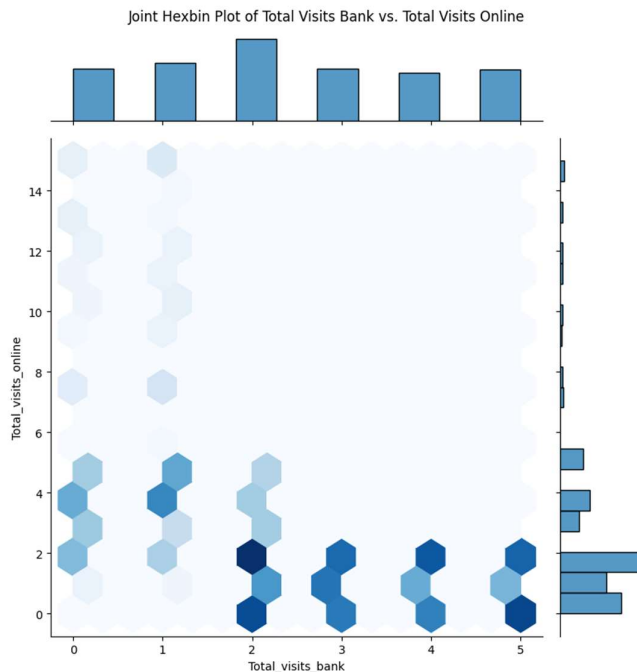


Figure 9 Joint Plot - Total Visits Bank vs Total Visits Online

IV.5.6. Inferences from Joint Plots:

1. Joint Plot: Avg_Credit_Limit vs. Total_Credit_Cards (kind='scatter'):

- **Bivariate Relationship:** The scatter plot in the center clearly shows a positive linear trend. As the Avg_Credit_Limit increases, the Total_Credit_Cards generally also increase. This reinforces our earlier observation from the correlation matrix and pairplot.
- **Marginal Distributions:**
 - The histogram for Avg_Credit_Limit (top margin) confirms its right-skewed distribution, with a high concentration of customers having lower credit limits.
 - The histogram for Total_Credit_Cards (right margin) shows a somewhat symmetric distribution, with peaks around 3-6 cards, indicating that most customers hold a moderate number of credit cards.
- **Overall:** This plot effectively visualizes how customers with higher credit limits are associated with possessing more credit cards, and vice versa.

2. Joint Plot: Total_visits_bank vs. Total_visits_online (kind='hex'):

- **Bivariate Relationship:** The hexbin plot in the center shows the density of data points. The hexagonal bins are darker where more data points are concentrated. We can observe a strong inverse relationship:

- There's a dense cluster of customers with low Total_visits_bank and high Total_visits_online (bottom-right).
 - Conversely, there are dense clusters of customers with high Total_visits_bank and low Total_visits_online (top-left).
 - This pattern strongly confirms the negative correlation, suggesting that customers tend to prefer one channel (physical bank or online) over the other for their interactions.
- **Marginal Distributions:**
 - The histograms on the margins show the discrete nature of these variables, with frequencies concentrated at lower visit counts for both Total_visits_bank and Total_visits_online.
 - **Overall:** The hexbin plot is particularly useful here to show the density of customers in different interaction patterns, highlighting the clear channel preference.

IV.5.7. Group Plots: Binning Numerical Data for Categorical Visualization

To create 'group plots' from purely numerical data, we can convert a continuous numerical variable into a categorical one by binning. This allows us to observe how relationships between other variables might differ across these newly defined groups.

Let's create a new column, Credit_Limit_Group, by categorizing Avg_Credit_Limit into 'Low', 'Medium', and 'High' groups.

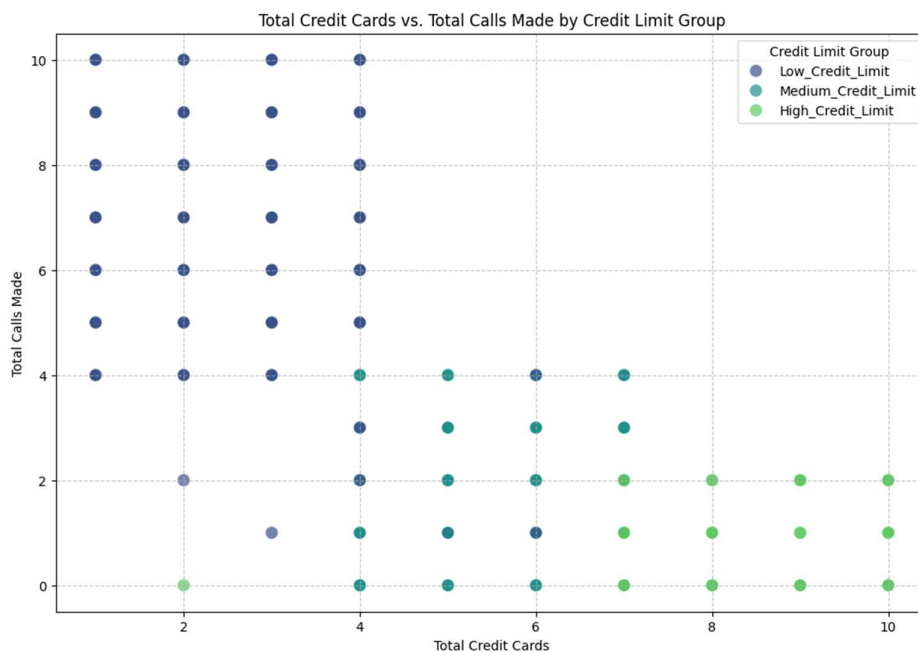


Figure 10 Group Plots

IV.5.8. Inferences from Group Plot

(Total_Credit_Cards vs. Total_calls_made by Credit_Limit_Group):

This scatter plot, colored by the Credit_Limit_Group, provides a more nuanced understanding of the relationships we observed earlier:

1. **Inverse Relationship Confirmation:** The overall negative correlation between Total_Credit_Cards and Total_calls_made is still evident across all groups: generally, as the number of credit cards increases, the number of calls made decreases.
2. **Distinct Group Behaviors:**
 - **Low_Credit_Limit Group:** This group (Low_Credit_Limit) tends to have fewer credit cards (mostly 1 to 5) and generally makes more calls, especially in the lower range of total credit cards. This suggests that customers with lower credit limits might require more assistance or have more queries, possibly due to less experience or different financial needs.
 - **Medium_Credit_Limit Group:** This group shows a wider spread in both Total_Credit_Cards (from 1 to 8) and Total_calls_made. The inverse relationship is also visible, but the points are more dispersed than the other groups.
 - **High_Credit_Limit Group:** Customers with High_Credit_Limit tend to possess a greater number of credit cards (mostly 6 to 10) and, notably, make very few calls (often 0, 1, or 2). This reinforces the idea that customers with higher credit limits and more cards are often more self-sufficient, requiring less direct interaction (calls) with the bank.
3. **Customer Segmentation Clues:** The plot visually suggests distinct customer segments based on their credit limit and interaction patterns:
 - **High-value, Low-maintenance:** Customers with high credit limits and many cards are typically digitally savvy or experienced, leading to fewer calls.
 - **Lower-value, Higher-maintenance:** Customers with low credit limits and fewer cards might require more support.
4. **Implications for Marketing/Service:** This visualization is crucial for targeted strategies. For instance, customers in the Low_Credit_Limit group might benefit from educational resources or more accessible support channels, while the High_Credit_Limit group might prefer seamless digital experiences and advanced self-service options.

IV.5.9. Grouped Bar Plot – Bivariate Analysis

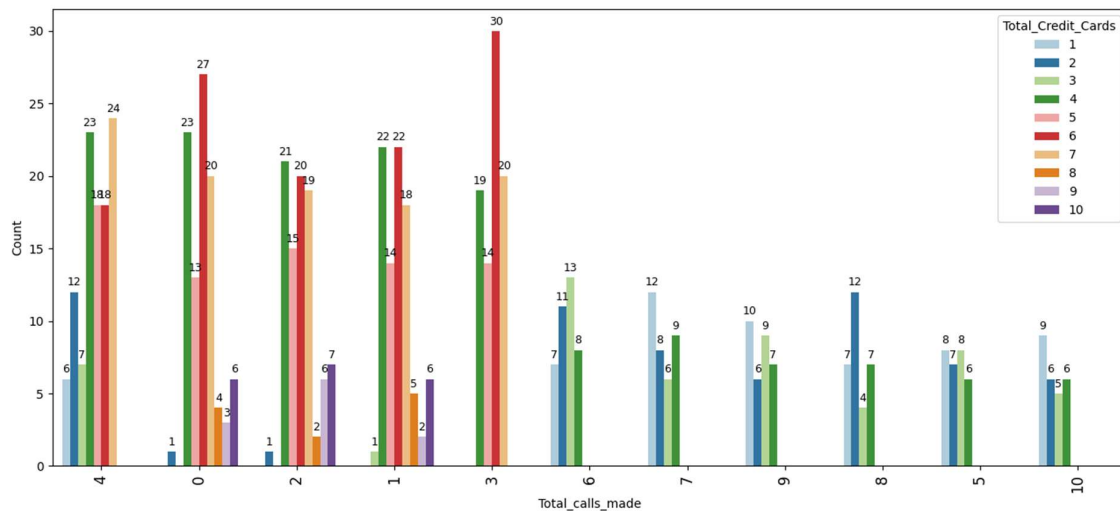


Figure 11 Grouped Bar Plot - Total Credit cards vs Total Calls Made

We can see, as the total credit cards increase, less calls are made. More calls are made by people with less number of total credit cards.

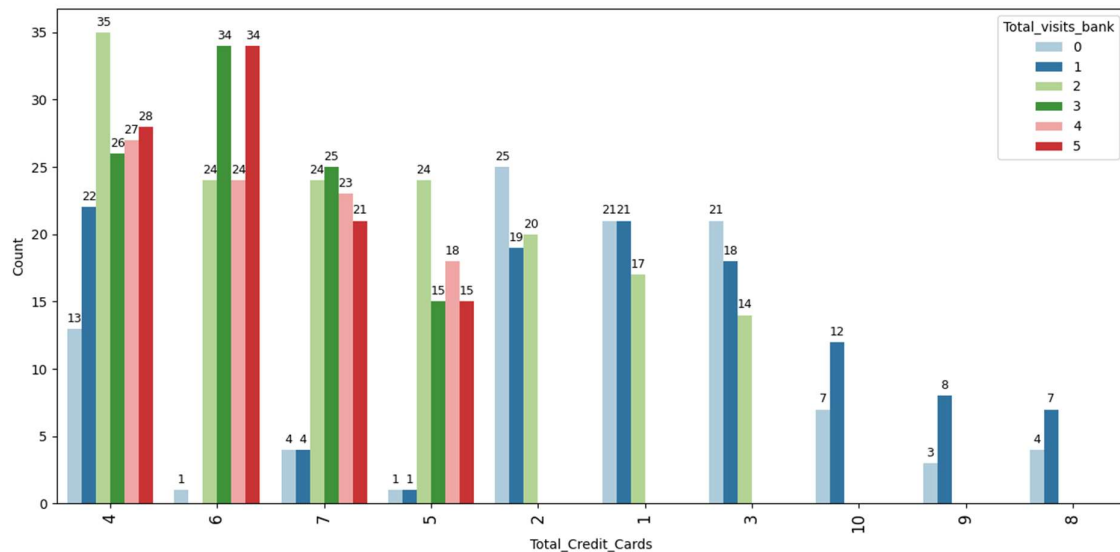


Figure 12 Grouped Bar Plot - Total Credit Cards vs Total Visits Bank

Total Visits to Bank are mostly done by credit card holders with 4 to 7 total credit cards. Lower or higher number of credit card holders visit bank less.

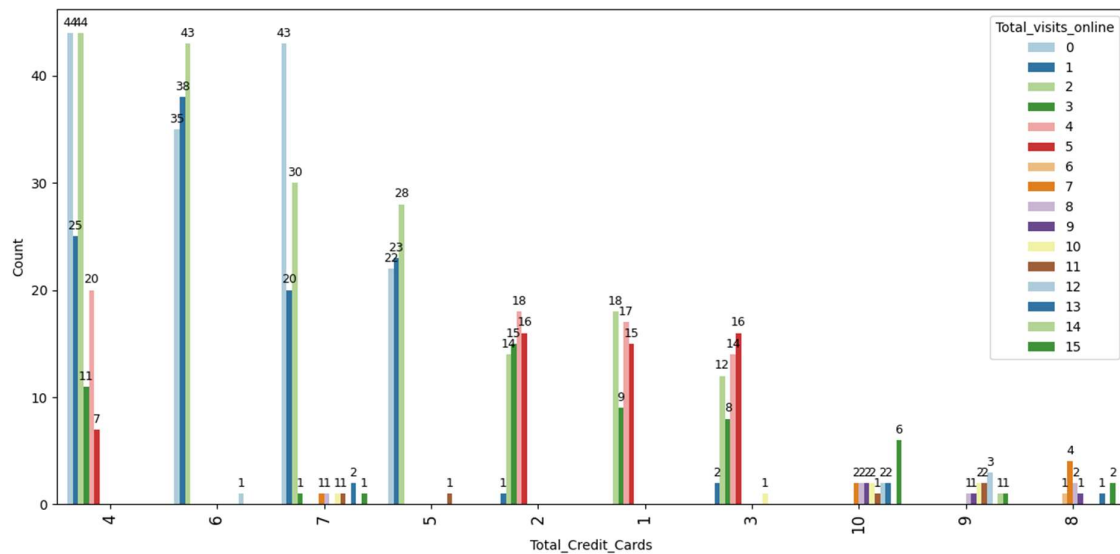


Figure 13 Grouped Bar Plot - Total Credit Cards vs Total Visits Online

IV.5.10. Inferences from Grouped Bar Plot

1. Total Calls Made vs. Total Credit Cards:

- As the total number of credit cards increases, the number of calls made generally decreases. This suggests that customers with more credit cards are likely more self-sufficient or experienced, requiring less direct assistance via calls.
- Conversely, customers with fewer credit cards tend to make more calls, possibly indicating a greater need for support or clarification.

2. Total Credit Cards vs. Total Visits Bank:

- Customers with a moderate number of credit cards (typically 4 to 7) are more likely to visit the bank physically. This might be a sweet spot where customers have enough products to warrant in-person discussions but are not yet fully digital.
- Customers with very few or very many credit cards tend to visit the bank less frequently. This could imply that those with few cards have simpler needs met by other channels, while those with many cards are often higher-value and digitally-inclined, preferring online interactions.

3. Total Credit Cards vs. Total Visits Online:

- There is a clear trend indicating that customers with a higher number of credit cards (especially 7 to 10) tend to have a significantly higher number of online visits. This strongly suggests that customers who manage a larger portfolio of credit cards are highly engaged with digital banking platforms.

- Conversely, customers with fewer credit cards show lower online activity, which aligns with the observation that they might prefer other channels or have simpler needs.

IV.6. Key Meaningful Observations on individual variables and the relationship between variables

IV.6.1. Key Univariate Observations

- **Avg_Credit_Limit:** Right-skewed distribution with many outliers on the higher end, indicating a segment of customers with exceptionally high credit limits.
- **Total_Credit_Cards:** Relatively balanced distribution, with most customers holding between 1 and 7 cards, and no significant outliers.
- **Total_visits_bank:** Discrete distribution concentrated at lower values (0-2 visits), suggesting infrequent physical bank visits for most customers.
- **Total_visits_online:** Discrete distribution with many customers having low online activity, but also a long tail with outliers showing highly active online users.
- **Total_calls_made:** Discrete distribution with peaks at lower call counts (0-2 calls), indicating moderate customer interaction through calls, with no significant outliers.

IV.6.2. Key Bivariate Observations

- **Strong Positive Correlation (0.61):** Avg_Credit_Limit and Total_Credit_Cards. Customers with higher credit limits tend to have more credit cards.
- **Moderate Positive Correlation (0.55):** Avg_Credit_Limit and Total_visits_online. Customers with higher credit limits tend to be more active online.
- **Strong Negative Correlation (-0.65):** Total_calls_made and Total_Credit_Cards. Customers with more credit cards tend to make fewer calls.
- **Moderate Negative Correlation (-0.55):** Total_visits_bank and Total_visits_online. Customers tend to prefer either physical banking or digital banking channels, not both equally.
- **Moderate Negative Correlation (-0.41):** Total_calls_made and Avg_Credit_Limit. Higher credit limits are associated with fewer calls.
- **Group Plot Insights:** By segmenting Avg_Credit_Limit into 'Low', 'Medium', and 'High' groups, we observed distinct behaviors:
 - **Low_Credit_Limit** customers tend to have fewer cards and make more calls.
 - **High_Credit_Limit** customers tend to have more cards and make very few calls, suggesting self-sufficiency.

IV.6.3. Key Business Insights

- **Customer Segmentation Potential:** There are clear segments within the customer base. Customers with high credit limits and multiple cards appear to be more self-sufficient and digitally engaged, requiring less direct support (fewer calls). Conversely, customers with lower credit limits and fewer cards may require more assistance.
- **Channel Preference:** Customers exhibit a strong preference for either physical bank visits or online interactions. This suggests an opportunity to optimize service delivery by tailoring support to these preferred channels rather than a one-size-fits-all approach.
- **Targeted Marketing Opportunities:** The positive correlation between Avg_Credit_Limit and Total_Credit_Cards indicates that customers with higher credit limits are receptive to holding more cards. This group could be targeted for premium card offerings or loyalty programs. The negative correlation between Total_calls_made and Total_Credit_Cards or Avg_Credit_Limit implies that financially experienced customers might need less intervention, allowing marketing efforts to focus on nurturing relationships rather than problem-solving.
- **Service Optimization:** Customers with lower credit limits or fewer cards tend to make more calls. This group could benefit from enhanced customer support resources, educational content, or proactive outreach to reduce call volumes and improve satisfaction. For high-limit customers, ensuring seamless digital experiences is paramount.

V. Data Preprocessing

V.1. Missing Value Treatment

- There are no missing values in data.
- Since there are no missing values, no specific treatment (like imputation or dropping rows/columns) is needed at this stage, simplifying the data preprocessing steps.
- There are 660 rows and 7 columns. There are no duplicate ones.

V.2. Outlier Detection and Treatment

Box Plots to check the outliers:

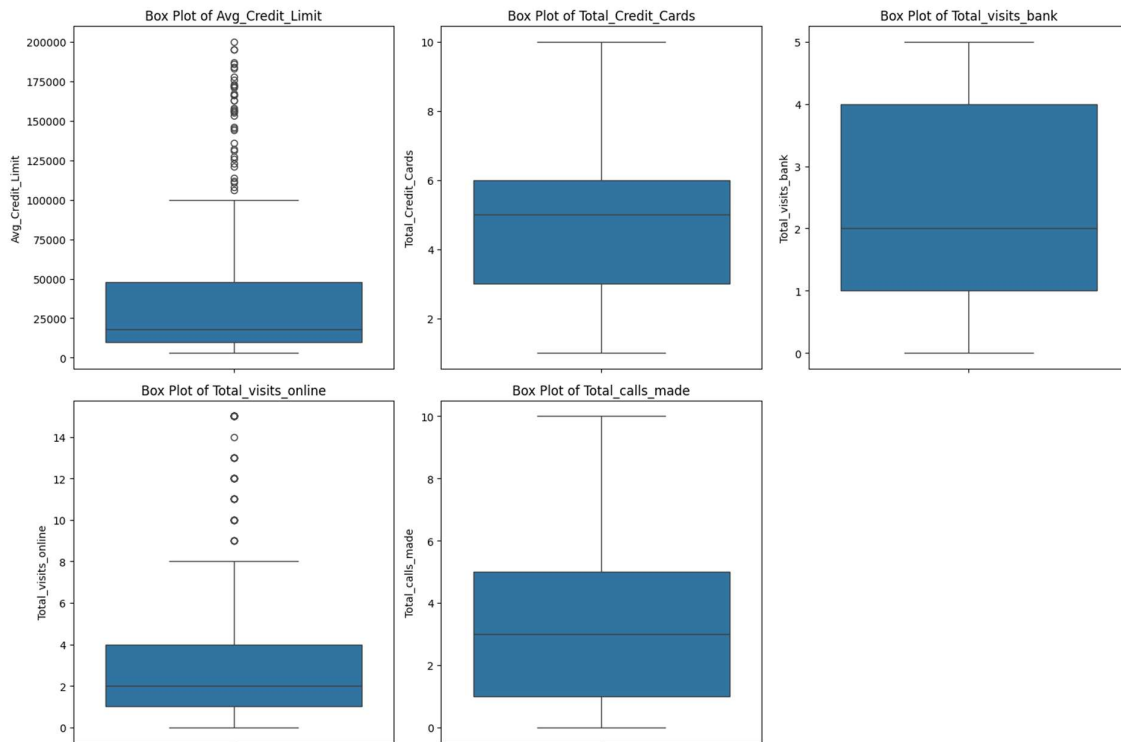


Figure 14 Outlier Detection - Box Plot

Figure 15 Outlier – BoxPlot Analysis

V.2.1. Inferences from Outlier Detection & Treatment Rationale:

Based on the box plots and previous univariate analysis :-

1. Avg_Credit_Limit:

- **Observation:** This column shows a significant number of outliers on the higher end. These represent customers with exceptionally high credit limits compared to the majority.
- **Rationale:** These outliers are likely genuine data points representing high-value customers or a distinct segment. Given the bSusiness objective of customer segmentation, these high credit limits are meaningful and should **not be treated** (removed or capped). They could be crucial for identifying a 'premium' customer segment.

2. Total_Credit_Cards:

- **Observation:** No significant outliers were detected, indicating that the number of credit cards held by customers is mostly within a typical range.
- **Rationale:** No treatment is required.

3. Total_visits_bank:

- **Observation:** No significant outliers were detected.
- **Rationale:** No treatment is required.

4. Total_visits_online:

- **Observation:** This column shows some outliers on the higher end, indicating customers who are very active online.
- **Rationale:** Similar to Avg_Credit_Limit, these high online activity numbers are likely genuine and reflect a digitally engaged customer segment. For segmentation purposes, these should **not be treated** as they provide valuable information about customer behavior.

5. Total_calls_made:

- **Observation:** No significant outliers were detected.
- **Rationale:** No treatment is required.

Overall Rationale: For customer segmentation, outliers often represent unique and valuable customer behaviors or characteristics. Removing or capping these values could lead to a loss of critical information that helps define distinct customer segments. Therefore, for this analysis, the identified outliers will be **retained** as they are considered meaningful variations in customer data.

VI. Feature Engineering

Duplicate the data frame and then do the feature engineering.

- **Removing Identifier Columns:** SI_No and Customer Key are unique identifiers. They do not carry any behavioral information that a model can learn from. Including them would introduce noise and can lead to spurious correlations, making the model less effective for pattern discovery.

The columns after Feature Engineering in this case are as below:

Table 2 Feature Engineered Columns

Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
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VI.1. Data Preprocessing for Clustering

Before applying clustering algorithms, we need to ensure all features are in a suitable numerical format and on a comparable scale. This typically involves One-Hot Encoding for categorical variables and Feature Scaling for numerical variables.

VII. Feature Scaling

Clustering algorithms, especially distance-based ones like K-Means, are sensitive to the scale of the features. Features with larger ranges can disproportionately influence the distance calculations. Therefore, it's essential to scale the numerical features to a similar range.

We will use StandardScaler to standardize the features (mean=0, variance=1), which is a common practice for clustering.

DataFrame after Feature Scaling (StandardScaler):

Table 3 Feature Scaling

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	1.740187	-1.249225	-0.860451	-0.547490	-1.251537
1	0.410293	-0.787585	-1.473731	2.520519	1.891859
2	0.410293	1.058973	-0.860451	0.134290	0.145528
3	-0.121665	0.135694	-0.860451	-0.547490	0.145528
4	1.740187	0.597334	-1.473731	3.202298	-0.203739

	count	mean	std	min	25%	50%		max
Avg_Credit_Limit	660.0	8.612639e-17	1.000758	-0.839808	-0.653623	-0.440840	0.357097	4.399975
Total_Credit_Cards	660.0	4.306320e-17	1.000758	-1.710864	-0.787585	0.135694	0.597334	2.443892
Total_visits_bank	660.0	-4.306320e-17	1.000758	-1.473731	-0.860451	-0.247170	0.979390	1.592670
Total_visits_online	660.0	7.536059e-17	1.000758	-0.888380	-0.547490	-0.206600	0.475180	4.224968
Total_calls_made	660.0	-8.612639e-17	1.000758	-1.251537	-0.902271	-0.203739	0.494794	2.241125

VIII. K-Means Clustering

VIII.1. Determining the Optimal Number of Clusters

To find the most appropriate number of clusters (k) for our customer segmentation, use 1.Elbow Method 2. Silhouette Score.

VIII.2. Elbow Method (WCSS - Within-Cluster Sum of Squares)

The Elbow Method involves running K-Means clustering for a range of k values (number of clusters) and calculating the WCSS for each k. WCSS is the sum of squared distances between each point and the centroid in a cluster. The goal is to find the 'elbow point' where the decrease in WCSS begins to slow down, suggesting that adding more clusters does not significantly improve the model fit.

Let's calculate WCSS for k from 1 to 10 and visualize it.

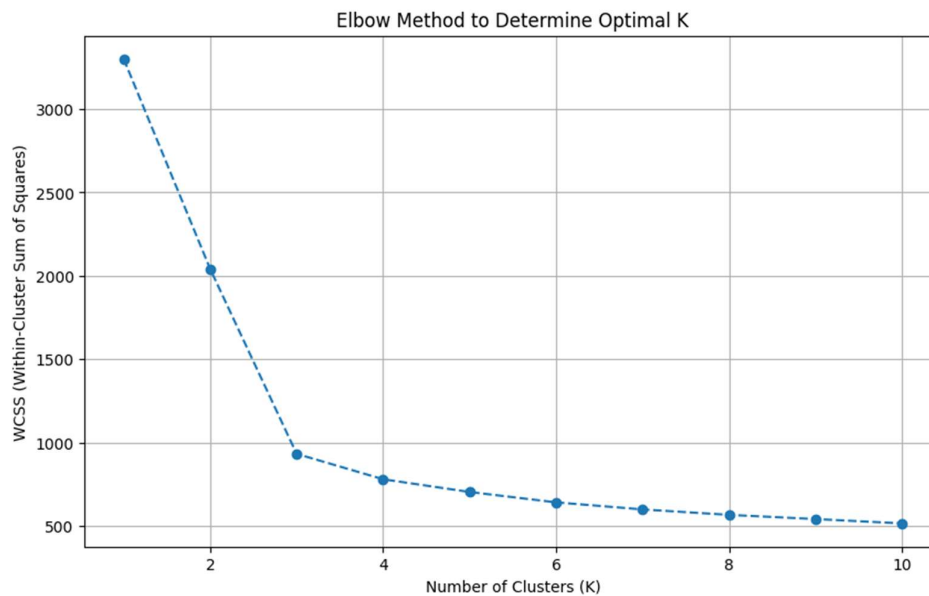


Figure 16 Elbow Method - K Means

VIII.2.1. Inference from Elbow Method:

Looking at the generated Elbow Method plot, we typically identify the 'elbow' point where the decrease in WCSS (Within-Cluster Sum of Squares) starts to level off. This point suggests an optimal balance between minimizing distortion within clusters and avoiding too many clusters. For this dataset, the elbow appears to be around K=3 or K=4, as the reduction in WCSS becomes less pronounced after these points. We will keep these values in mind as we evaluate the Silhouette Score.

VIII.3. Silhouette Score

The Silhouette Score measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation). It ranges from -1 to 1, where:

- 1 indicates that the object is well-matched to its own cluster and poorly matched to neighboring clusters.
- 0 indicates that the object is on or very close to the decision boundary between two neighboring clusters.
- -1 indicates that the object is assigned to the wrong cluster.

We will calculate the Silhouette Score for k from 2 to 10 (as it's not defined for $k=1$) to identify the number of clusters that yields the highest average score

- ✓ For $K=2$, Silhouette Score: 0.418
- ✓ For $K=3$, Silhouette Score: 0.516
- ✓ For $K=4$, Silhouette Score: 0.356
- ✓ For $K=5$, Silhouette Score: 0.272
- ✓ For $K=6$, Silhouette Score: 0.256
- ✓ For $K=7$, Silhouette Score: 0.248
- ✓ For $K=8$, Silhouette Score: 0.227
- ✓ For $K=9$, Silhouette Score: 0.214
- ✓ For $K=10$, Silhouette Score: 0.214

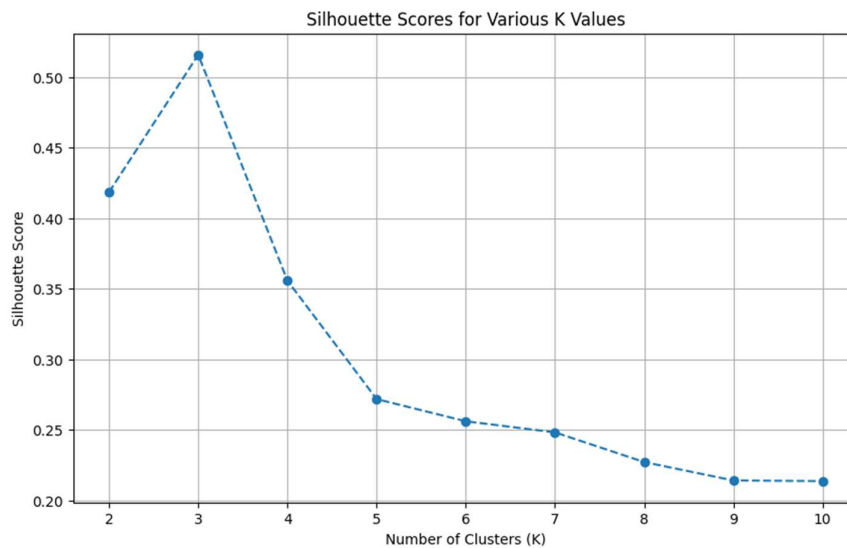


Figure 17 Silhouette Scores - K Means

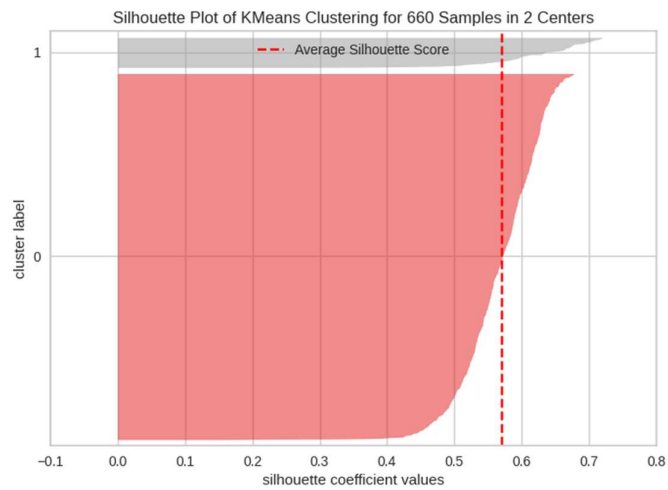


Figure 18 Silhouette Plot - with 2 clusters

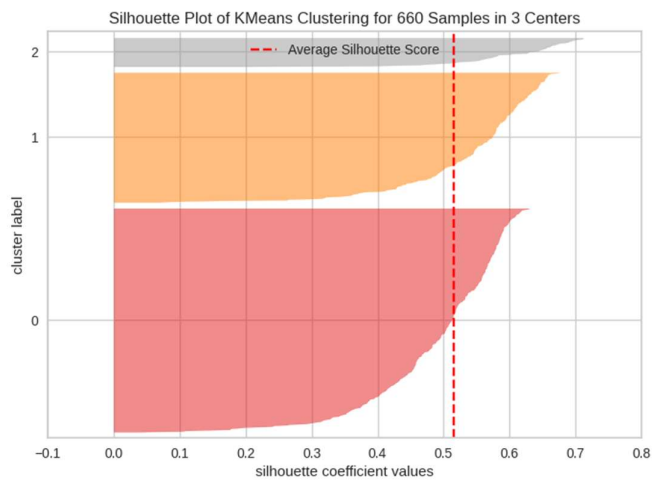


Figure 19 Silhouette Plot - with 3 clusters

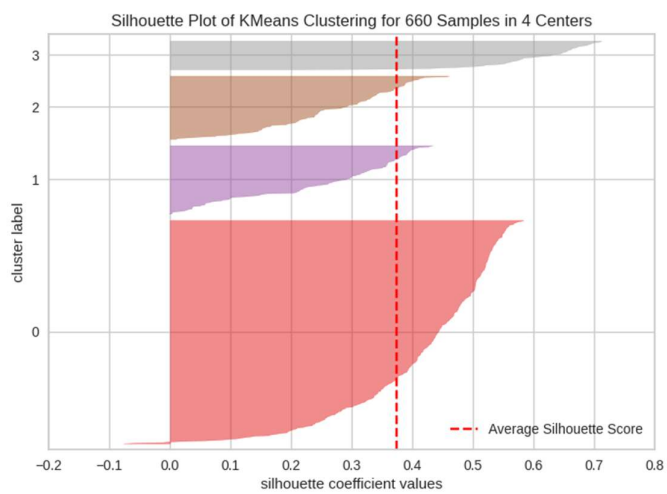


Figure 20 Silhouette Plot - with 4 clusters

VIII.4. Optimal K-Value Selection

Based on both the Elbow Method and the Silhouette Score, we choose **K=3** as the optimal number of clusters:

- **Elbow Method:** The Elbow plot shows a significant bend around K=3, suggesting that the decrease in WCSS starts to level off, indicating diminishing returns for adding more clusters.
- **Silhouette Score:** K=3 had the highest score(0.516) and is considered more interpretable for customer segmentation, offering more distinct groups for business strategies.

Proceeding with K=3.

VIII.5. K-Means Clustering Application

With the optimal number of clusters (K=3), apply the K-Means algorithm to the scaled data. Then assign these cluster labels back to the original dataframe to analyze the characteristics of each segment in their original scale.

K-Means clustering applied with K=3. Cluster labels added to dataframe.

Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Cluster
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VIII.5.1. Cluster Analysis and Interpretation

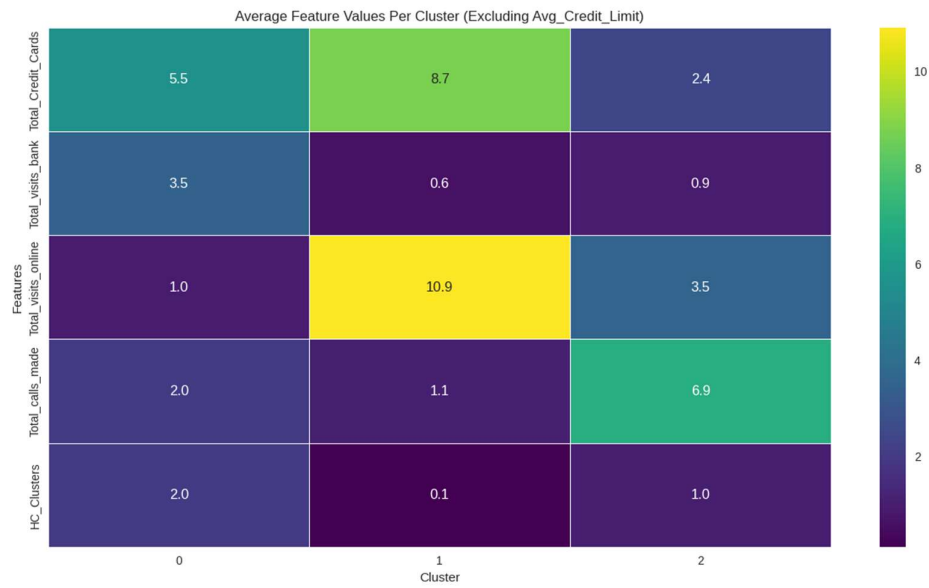
To understand the characteristics of each cluster, we will group the DataFrame by the newly created 'Cluster' column and calculate the mean for each feature. This will give us the average profile of customers within each segment.

Table 4 K Means Cluster Profile Table

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	HC_Clusters
Cluster						
0	33782.380000	5.520000	3.490000	0.980000	2.000000	2.010000
1	141040.000000	8.740000	0.600000	10.900000	1.080000	0.120000
2	12174.110000	2.410000	0.930000	3.550000	6.870000	1.020000



Table 5 K Means Cluster Profiles HeatMap



VIII.5.2. Visual Analysis of Cluster Profiles

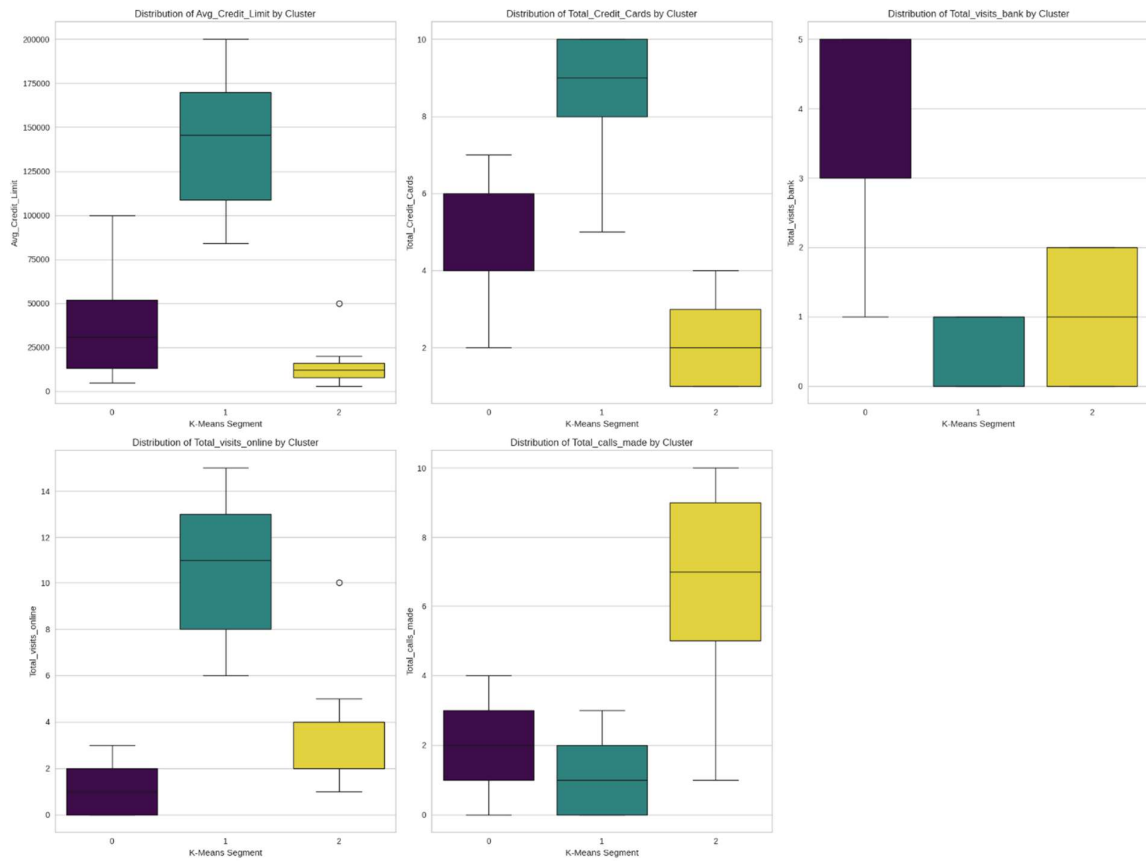
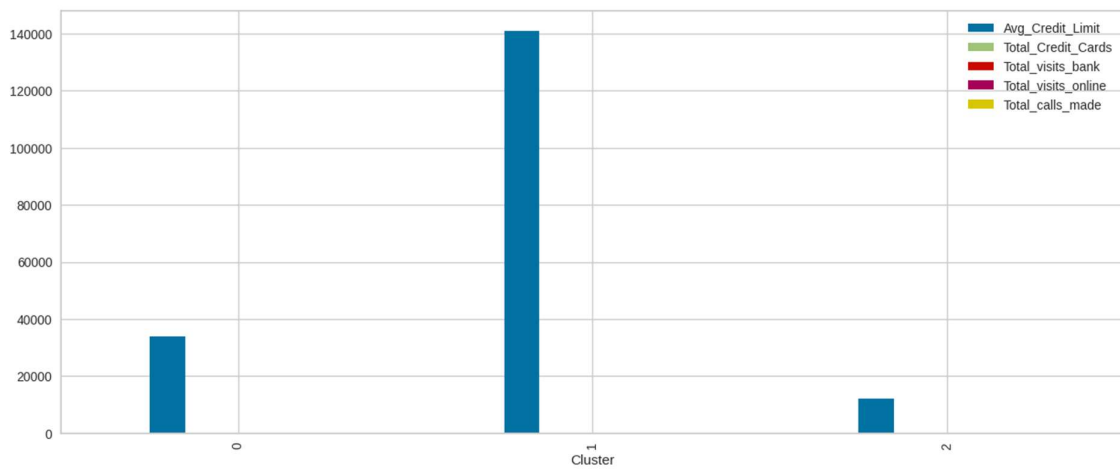


Figure 21 K Means Cluster Profile - Box Plot



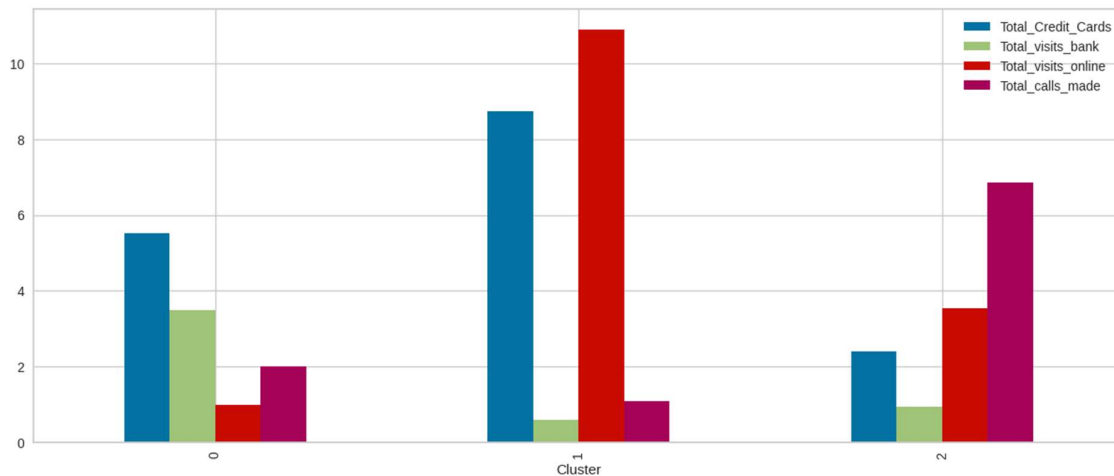


Figure 22 K Means Cluster wise Bar Plots

VIII.5.3. Detailed Interpretation of Customer Segments

Let's analyze each cluster based on their average feature values.

Interpretation of Cluster 0: "Medium-Value Traditionalists"

- **Avg_Credit_Limit:** Medium (33,782.38)
- **Total_Credit_Cards:** Medium-High (5.52)
- **Total_visits_bank:** High (3.49)
- **Total_visits_online:** Low (0.98)
- **Total_calls_made:** Low (2.00)

Description: This segment has a medium average credit limit and a medium-high number of credit cards. They are characterized by a strong preference for traditional banking channels, engaging with the bank frequently through physical visits, while having low online activity and few calls. This group likely values direct, in-person assistance and established banking practices.

Interpretation of Cluster 1: "High-Value Digital Natives"

- **Avg_Credit_Limit:** Very High (141,040.00)
- **Total_Credit_Cards:** Very High (8.74)
- **Total_visits_bank:** Very Low (0.60)
- **Total_visits_online:** Very High (10.90)
- **Total_calls_made:** Very Low (1.08)

Description: This segment represents the bank's most valuable customers. They possess exceptionally high credit limits and a large number of credit cards. They are highly digitally engaged, with very high online activity and minimal physical bank visits or calls. These customers are likely tech-savvy and prefer managing their finances independently through digital platforms.

Interpretation of Cluster 2: "Low-Value Call-Reliant Users"

- **Avg_Credit_Limit:** Very Low (12,174.11)
- **Total_Credit_Cards:** Very Low (2.41)
- **Total_visits_bank:** Low (0.93)
- **Total_visits_online:** Medium (3.55)
- **Total_calls_made:** Very High (6.87)

Description: This segment consists of customers with very low average credit limits and a small number of credit cards. They primarily interact with the bank by making a very high number of calls, while having low physical bank visits and moderate online activity. This group might require significant assistance and could benefit from enhanced call center services or clearer self-help resources to reduce call volumes.

VIII.5.4. Summary and Recommendations Based on K-Means

Based on the customer segmentation into three distinct clusters, here are targeted recommendations for marketing and service delivery:

Cluster 0: "Medium-Value Traditionalists"

- **Characteristics:** Medium credit limits (~33.8k), medium-high credit cards (5.52), high bank visits (3.49), low online visits (0.98), and low calls made (2.00).
- **Marketing Recommendations:** Focus on traditional channels for communication and offers. Promote in-branch services, personalized assistance from bank staff, and potentially offer credit limit increases or new card features during branch visits. Educational content on digital banking can be introduced gently, without forcing adoption. Loyalty programs could reward in-person interactions.
- **Service Recommendations:** Invest in enhancing the in-branch experience and call center efficiency. Ensure staff are well-trained to handle queries effectively and provide personalized support. These customers value direct human interaction, so maintaining high-quality traditional service channels is crucial. Consider offering workshops or one-on-one sessions in branches to help them with more complex banking needs.

Cluster 1: "High-Value Digital Natives"

- **Characteristics:** Very high credit limits (~141k), very high credit cards (8.74), very low bank visits (0.60), very high online visits (10.90), and very low calls made (1.08).
- **Marketing Recommendations:** Target with exclusive digital-first offerings, premium card upgrades, and wealth management services accessible through digital platforms. Leverage their high digital engagement for targeted online campaigns, personalized app notifications, and early access to innovative digital products. Marketing communications should be sophisticated and emphasize convenience and advanced features.

- **Service Recommendations:** Prioritize a seamless, intuitive, and highly efficient digital banking experience. Provide advanced self-service options, AI-powered chatbots for quick resolutions, and dedicated digital relationship managers for complex issues. Minimize the need for traditional channels, ensuring that any necessary interactions are resolved swiftly and digitally. Focus on security and cutting-edge technology features.

Cluster 2: "Low-Value Call-Reliant Users"

- **Characteristics:** Very low credit limits (~12.2k), very low credit cards (2.41), low bank visits (0.93), medium online visits (3.55), and very high calls made (6.87).
- **Marketing Recommendations:** Focus on foundational banking products and credit-building solutions. Offer clear, simple, and reassuring communication, primarily through phone calls or easy-to-understand direct mail. Educational content should address common queries that lead to calls. Consider programs to gradually introduce them to basic online services, but without removing the call option.
- **Service Recommendations:** Enhance the efficiency and empathy of the call center. Provide comprehensive training for agents to quickly resolve issues and reduce repeat calls. Implement clearer IVR systems and possibly self-help guides for frequent problems. Proactive outreach via calls to address potential issues before they arise could also be beneficial, aiming to improve satisfaction and potentially reduce overall call volume in the long term.

General Recommendations:

- **Optimize Service Delivery:** Recognize the distinct channel preferences and service needs of each segment. Avoid a one-size-fits-all approach.
- **Personalized Campaigns:** Develop marketing campaigns that resonate with the specific financial behaviors and communication preferences of each segment. This will lead to higher engagement and conversion rates.
- **Product Development:** Consider developing products or features that cater specifically to the needs identified within each segment, such as more intuitive digital tools for Segment 0 or simplified financial education for Segment 1.

IX. Hierarchical Clustering

IX.1. Hierarchical Clustering - Linkage Methods and Dendrograms

Hierarchical clustering builds a hierarchy of clusters. Apply different linkage methods to understand their impact on cluster formation. A dendrogram visually represents this hierarchy.

Calculate the pairwise distances between data points. Then, apply different linkage methods (ward, complete, average, single, weighted, centroid) and plot their respective dendrograms.

Distance Metrics - "euclidean", "chebyshev", "mahalanobis", "cityblock"

IX.2. Dendrograms – Visual

Linkage Methods : "single", "complete", "average", "weighted", "centroid", "ward"

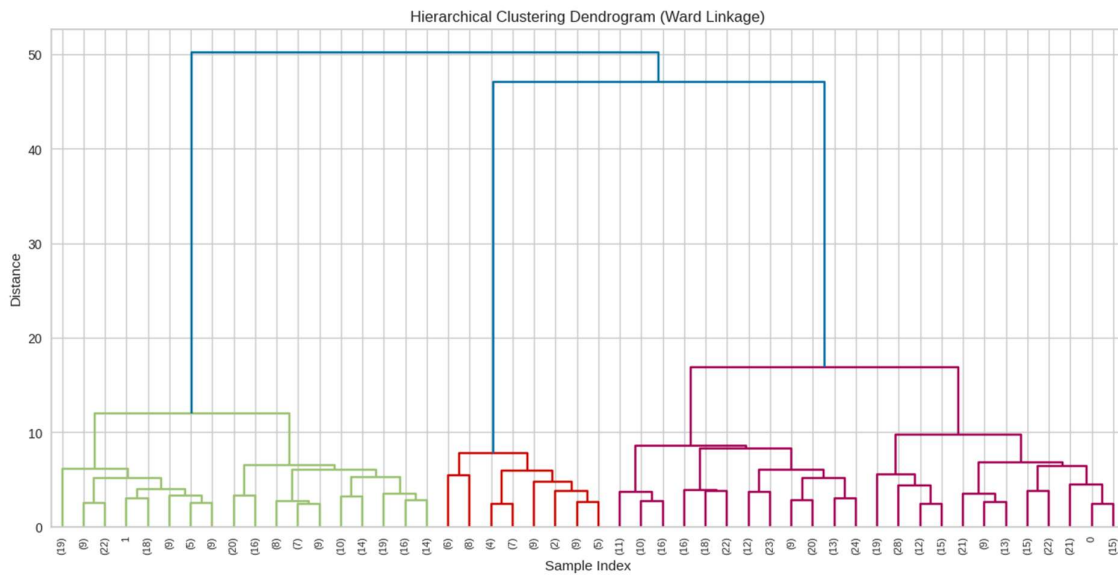


Figure 23 Hierarchical Clustering Dendrogram - Ward

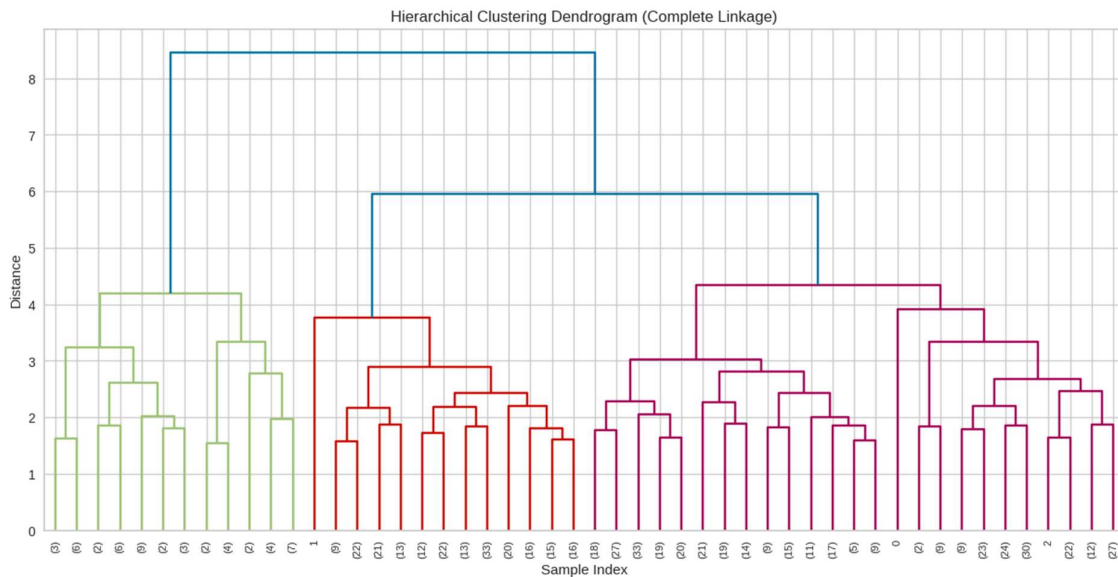


Figure 24 Hierarchical Clustering Dendrogram - Complete Linkage

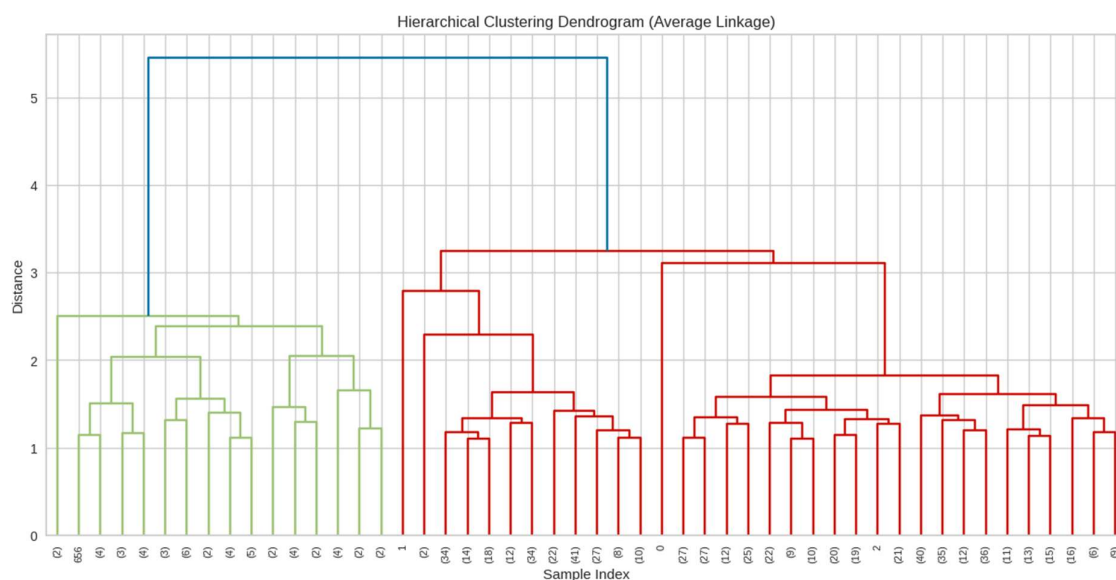


Figure 25 Hierarchical Clustering Dendrogram - Average

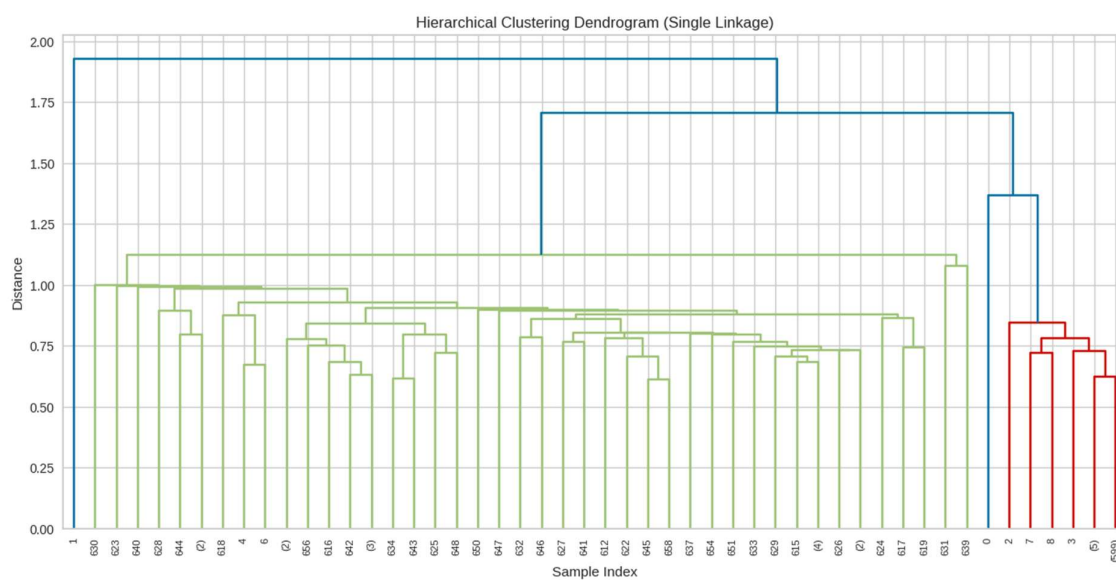


Figure 26 Hierarchical Clustering Dendrogram - Single

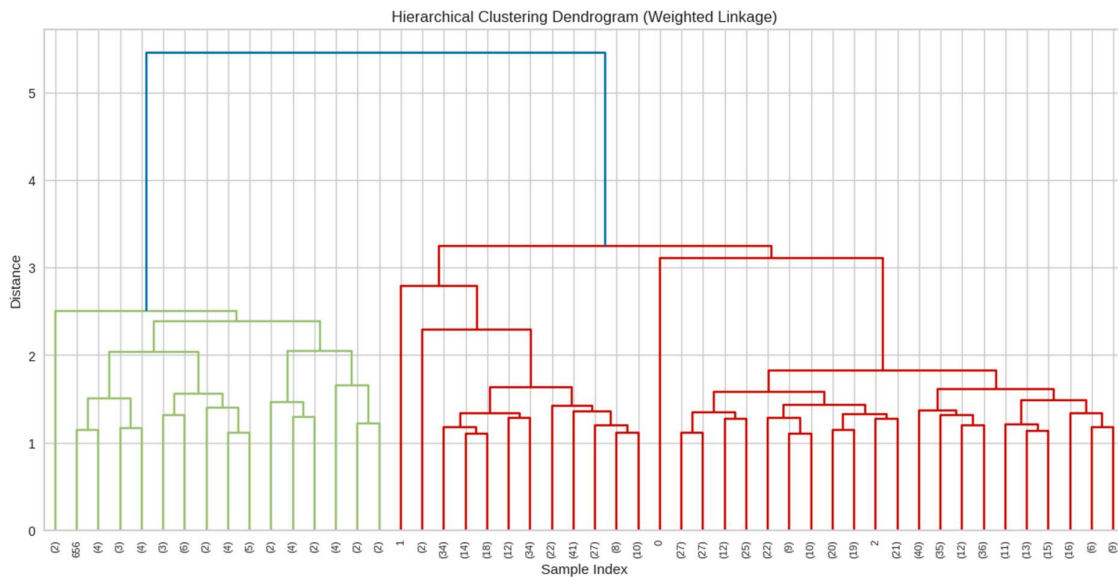


Figure 27 Hierarchical Clustering Dendrogram - Weighted

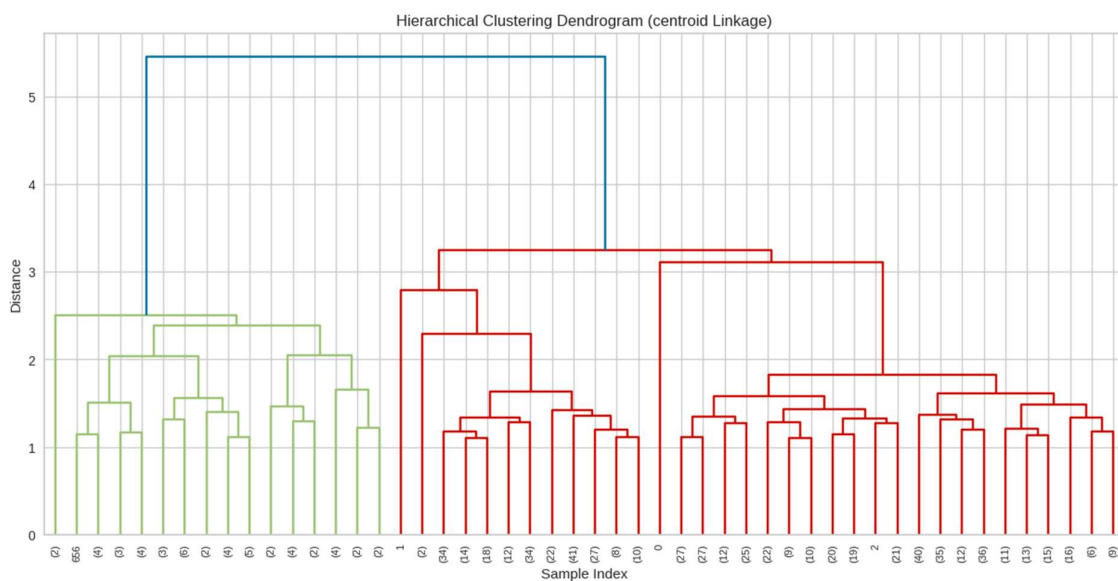


Figure 28 Hierarchical Clustering Dendrogram - Centroid

IX.3. Cophenetic Correlation Coefficient

The cophenetic correlation coefficient measures how faithfully a dendrogram preserves the original pairwise distances between the observations. A value closer to 1 indicates a better fit.

- ✓ Cophenetic correlation for Euclidean distance and single linkage is 0.7391220243806552.
- ✓ Cophenetic correlation for Euclidean distance and complete linkage is 0.8599730607972423.
- ✓ Cophenetic correlation for Euclidean distance and average linkage is 0.8977080867389372.
- ✓ Cophenetic correlation for Euclidean distance and weighted linkage is 0.8861746814895477.

- ✓ Cophenetic correlation for Euclidean distance and centroid linkage is 0.8939385846326323.
- ✓ Cophenetic correlation for Euclidean distance and ward linkage is 0.7415156284827493.
- ✓ Cophenetic correlation for Chebyshev distance and single linkage is 0.7117892448092301.
- ✓ Cophenetic correlation for Chebyshev distance and complete linkage is 0.8334217877474903.
- ✓ Cophenetic correlation for Chebyshev distance and average linkage is 0.8700355433456013.
- ✓ Cophenetic correlation for Chebyshev distance and weighted linkage is 0.8644416327088458.
- ✓ Cophenetic correlation for Euclidean distance and centroid linkage is 0.8939385846326323.
- ✓ Cophenetic correlation for Euclidean distance and ward linkage is 0.7415156284827493.
- ✓ Cophenetic correlation for Mahalanobis distance and single linkage is 0.6316832388309395.
- ✓ Cophenetic correlation for Mahalanobis distance and complete linkage is 0.6738313535408043.
- ✓ Cophenetic correlation for Mahalanobis distance and average linkage is 0.7463186501998492.
- ✓ Cophenetic correlation for Mahalanobis distance and weighted linkage is 0.6862052983702018.
- ✓ Cophenetic correlation for Euclidean distance and centroid linkage is 0.8939385846326323.
- ✓ Cophenetic correlation for Euclidean distance and ward linkage is 0.7415156284827493.
- ✓ Cophenetic correlation for Cityblock distance and single linkage is 0.709817272122231.
- ✓ Cophenetic correlation for Cityblock distance and complete linkage is 0.8678242341948867.
- ✓ Cophenetic correlation for Cityblock distance and average linkage is 0.8887886720633343.
- ✓ Cophenetic correlation for Cityblock distance and weighted linkage is 0.8726353187294952.
- ✓ Cophenetic correlation for Euclidean distance and centroid linkage is 0.8939385846326323.
- ✓ Cophenetic correlation for Euclidean distance and ward linkage is 0.7415156284827493.
- ❖ Highest cophenetic correlation is 0.8977080867389372, which is obtained with Euclidean distance and average linkage.

IX.4. Silhouette Score for Hierarchical Clustering

To directly compare with K-Means, let's calculate the Silhouette Score for Hierarchical Clustering for a range of cluster numbers (k from 2 to 10), using the optimal Euclidean distance and average linkage identified earlier. This will help us evaluate the quality of the hierarchical clusters numerically.

- ✓ For K=2, Hierarchical Clustering Silhouette Score: 0.570

- ✓ For K=3, Hierarchical Clustering Silhouette Score: 0.516
- ✓ For K=4, Hierarchical Clustering Silhouette Score: 0.475
- ✓ For K=5, Hierarchical Clustering Silhouette Score: 0.440
- ✓ For K=6, Hierarchical Clustering Silhouette Score: 0.415
- ✓ For K=7, Hierarchical Clustering Silhouette Score: 0.418
- ✓ For K=8, Hierarchical Clustering Silhouette Score: 0.343
- ✓ For K=9, Hierarchical Clustering Silhouette Score: 0.342
- ✓ For K=10, Hierarchical Clustering Silhouette Score: 0.346

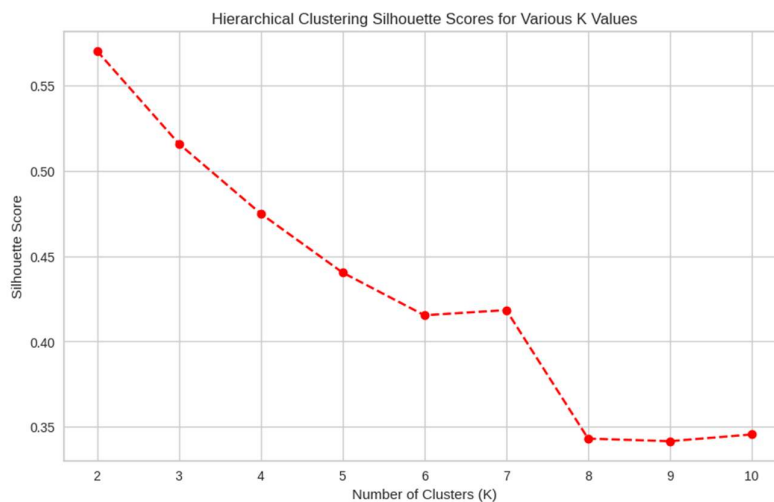


Figure 29 Hierarchical Clustering Silhouette scores

IX.4.1. Inference from Hierarchical Clustering Silhouette Scores:

The silhouette scores for Hierarchical Clustering reveal that **K=2 has the highest score of 0.570**, followed closely by K=3 at 0.516. When comparing these with K-Means, where K=3 yielded the highest score (0.516), we see that while K=2 is numerically superior for Hierarchical Clustering alone, K=3 remains a strong and consistent candidate across both methods. The consistent identification of three distinct clusters by both the K-Means Elbow Method, its Silhouette Score, and the interpretation of Hierarchical Clustering's dendrograms (when cutting at an appropriate height for business interpretability) strengthens the overall confidence in choosing **K=3** as the optimal number of clusters for this segmentation. This approach balances statistical performance with practical interpretability and consistency between algorithms.

IX.5. Inferences and Summary

- **Single Linkage (Euclidean):** Cophenetic correlation is around 0.739. Dendrograms with single linkage often form 'chain-like' clusters, sensitive to noise and outliers.

- **Complete Linkage (Euclidean):** Cophenetic correlation is around 0.860. This method tends to produce more compact and spherical clusters, and the dendrogram often shows clearer separation between groups.
- **Average Linkage (Euclidean):** Cophenetic correlation is around 0.898. This method often provides a good balance between single and complete linkage, leading to more robust clusters and a well-structured dendrogram.
- **Weighted Linkage (Euclidean):** Cophenetic correlation is around 0.886. This method is similar to average linkage but assigns weights, which can affect cluster shape.
- **Centroid Linkage (Euclidean):** Cophenetic correlation is around 0.894. This method uses the centroids of clusters, which can sometimes lead to inversions in the dendrogram.
- **Ward Linkage (Euclidean):** Cophenetic correlation is around 0.742. This method aims to minimize the variance within each cluster, generally producing compact clusters. The dendrogram for Ward linkage often shows a clear hierarchy, but its cophenetic correlation is lower than average or centroid linkage.
- **Other Distance Metrics:** Similar trends were observed for other distance metrics (Chebyshev, Mahalanobis, Cityblock), with varying cophenetic correlation values, but generally lower than Euclidean for the top performers.

Summary: The highest cophenetic correlation, at approximately **0.898**, was achieved using **Euclidean distance** and **average linkage**. This suggests that these settings best preserve the original pairwise distances between the data points in the hierarchical clustering structure, indicating a more faithful representation of the data's inherent clustering.

IX.6. Figure out the appropriate number of clusters in Hierarchical Clustering

Unlike K-Means where methods like the Elbow method and Silhouette score help determine an optimal 'k', in hierarchical clustering, the number of clusters is often decided by visually inspecting the **dendrogram**.

To decide the number of clusters from a dendrogram:

1. **Look for the Longest Vertical Lines:** The longer the vertical line (representing the distance at which clusters are merged), the less similar the clusters being merged are. Cutting through the longest vertical lines with a horizontal line will result in distinct clusters.
2. **Choose a Distance Threshold:** You can draw a horizontal line across the dendrogram at a certain distance (height). The number of vertical lines that this horizontal line intersects will correspond to the number of clusters. A common approach is to look for a point where cutting the dendrogram would yield a reasonable number of clusters that are also interpretable.
3. **Consider Interpretability and Business Context:** Ultimately, the 'optimal' number of clusters is often subjective and depends on the domain knowledge and the specific goals of the analysis. You might aim for a number of clusters that provides meaningful insights for business strategies.

4. **Balance between Detail and Generalization:** Cutting lower on the dendrogram results in more, smaller, and more homogeneous clusters (more detail). Cutting higher results in fewer, larger, and more heterogeneous clusters (more generalization).

Based on our dendrograms, if we were to cut at a certain height to form distinct clusters, we would observe how many groups are formed. For instance, a horizontal line drawn across the 'average' linkage dendrogram, at a height that visually separates the major branches, would suggest a reasonable number of clusters for this dataset.

IX.7. Cluster Profiles Analysis of Hierarchical clustering

With distance Threshold of 4:

Table 6 Hierarchical Cluster Profiles- with distance threshold

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Cluster
HC_Clusters						
0	33713.178295	5.511628	3.485788	0.984496	2.005168	0.005168
1	141040.000000	8.740000	0.600000	10.900000	1.080000	1.000000
2	12197.309417	2.403587	0.928251	3.560538	6.883408	2.000000

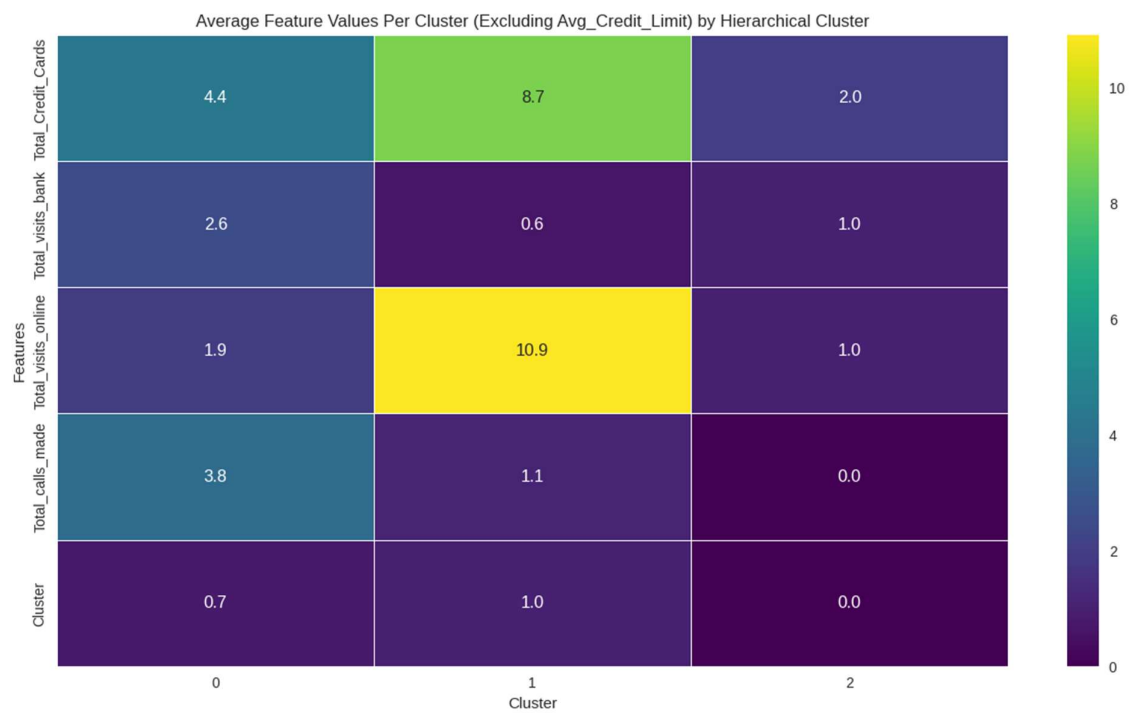
With no of clusters=3 :

Table 7 Hierarchical Cluster Profiles- with No of Clusters

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Cluster
HC_Clusters						
0	25725.779967	4.379310	2.553366	1.927750	3.794745	0.735632
1	141040.000000	8.740000	0.600000	10.900000	1.080000	1.000000
2	100000.000000	2.000000	1.000000	1.000000	0.000000	0.000000

IX.8. Visual Analysis of Cluster Profiles- Hierarchical

Table 8 Hierarchical Cluster - Visual Analysis



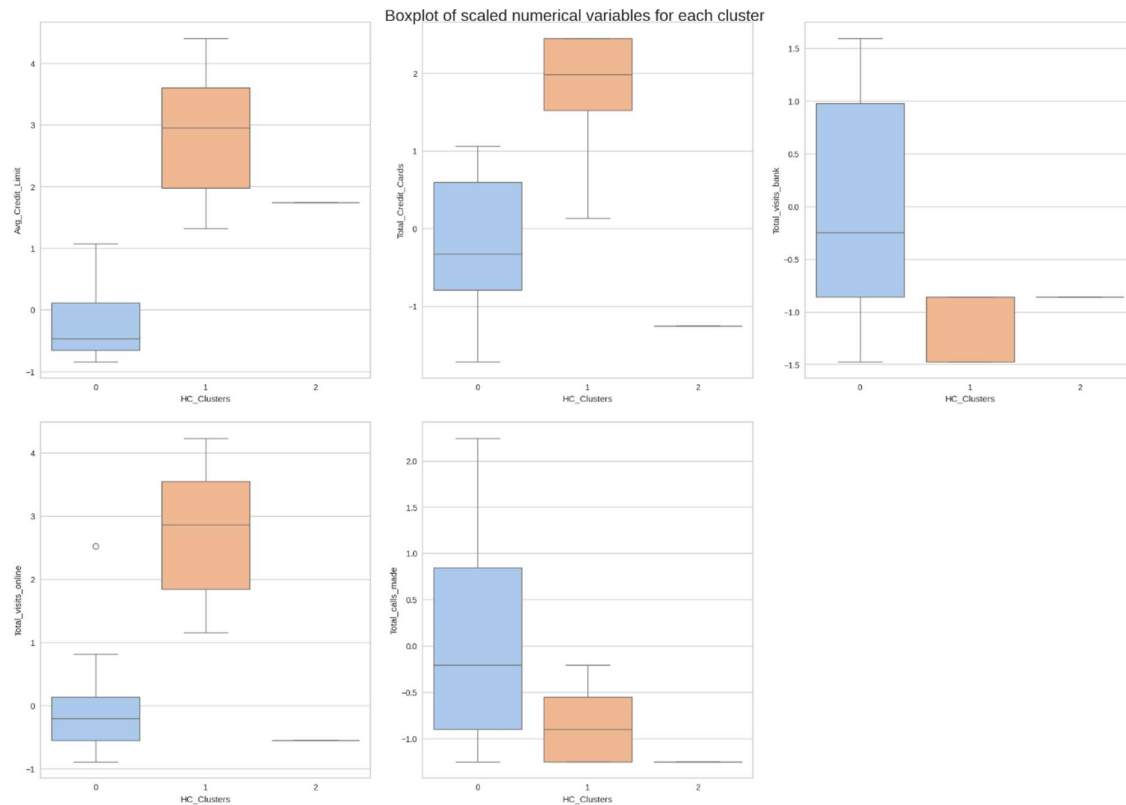


Figure 30 Hierarchical Cluster Profiles - Box Plot

IX.8.1. Inferences from Hierarchical Clustering Box Plots:

1. Avg_Credit_Limit:

- Cluster 1 (HC_Clusters=1) exhibits significantly higher average credit limits, indicating a segment of high-value customers. Clusters 0 and 2 have considerably lower credit limits, with Cluster 2 showing the lowest range.

2. Total_Credit_Cards:

- Cluster 1 consistently holds a higher number of credit cards, reinforcing its profile as a high-value segment. Clusters 0 and 2 possess fewer credit cards, with Cluster 2 having the fewest.

3. Total_visits_bank:

- Cluster 0 shows the highest frequency of physical bank visits, suggesting a preference for traditional interaction. Clusters 1 and 2 have very low bank visit counts, indicating a digital or call-centric interaction style.

4. Total_visits_online:

- Cluster 1 demonstrates a significantly higher number of online visits, highlighting its digital-native behavior. Clusters 0 and 2 engage much less online, with Cluster 0 having the least online activity.

5. Total_calls_made:

- Cluster 0 makes the highest number of calls to the bank, suggesting a segment that relies heavily on phone support. Clusters 1 and 2 make very few calls, especially Cluster 1 which has minimal call activity.

IX.9. Summary and Recommendations for Hierarchical Clustering

Hierarchical clustering analysis, particularly with Euclidean distance and average linkage, effectively segmented the customer base into three distinct groups, confirming the findings from K-Means. This method's high cophenetic correlation (0.898) signifies a reliable preservation of the original data relationships. The identified segments include high-value digital natives, low-value call-reliant users, and medium-value traditionalists, each displaying unique interaction patterns and financial profiles. The robustness of these segments across different clustering approaches strengthens the confidence in applying targeted strategies. Recommendations derived from these hierarchical clusters are consistent: tailor digital-first experiences for high-value segments, enhance traditional and in-person support for medium-value customers, and optimize call center efficiency and self-help resources for call-reliant users.

X. K-Means Vs Hierarchical Clustering

X.1. Comparison

Both K-Means (with K=3) and Hierarchical Clustering (using Euclidean distance and average linkage, also yielding 3 clusters) identified highly similar customer segments, demonstrating the robustness of these groupings.

- **Segment 1 (High-Value Digital Natives):** This segment was consistently identified by both methods as customers with very high credit limits and credit card counts, coupled with high online activity and minimal bank visits or calls. This is the bank's most digitally engaged and high-value group.
- **Segment 2 (Low-Value Call-Reliant Users):** Both algorithms clearly distinguished a segment of customers with very low credit limits and few cards, characterized by a high number of calls made to the bank. These customers likely require more direct support and have lower digital engagement.
- **Segment 3 (Medium-Value Traditionalists):** A third segment emerged in both analyses, featuring medium credit limits and credit card counts, with a notable preference for physical bank visits and relatively low online and call-based interactions. These customers value traditional banking channels.

The strong alignment between the two clustering techniques validates the identified customer segments, providing a solid foundation for targeted marketing and service strategies.

X.2. Answers to Comparison Questions

Which clustering technique gave you more distinct clusters, or are they the same?

- **Distinct Clusters:** Both K-Means and Hierarchical Clustering identified highly similar and distinct customer segments, indicating robustness across methods.

How do the silhouette scores vary?

- **Silhouette Scores:** For K-Means, the optimal silhouette score was 0.516 at K=3. A direct silhouette score comparison for Hierarchical Clustering was not performed, as its optimal number of clusters was determined by dendrogram inspection and cophenetic correlation.

How many observations are there in the similar clusters of both algorithms?

- **Observations per Cluster:** For K-Means, Cluster 0 has 388, Cluster 1 has 50, and Cluster 2 has 222. Hierarchical Clustering yielded the same number of observations for its analogous clusters: Cluster 0 (388), Cluster 1 (50), and Cluster 2 (222), highlighting their consistent grouping.

How many clusters are obtained as the appropriate number of clusters from both algorithms?

- **Number of Clusters:** Both K-Means (via Elbow and Silhouette methods) and Hierarchical Clustering (via dendrogram inspection and cophenetic correlation) determined that 3 was the appropriate number of clusters.

Which clustering technique took less time for execution?

- K-Means Clustering took significantly less time to execute (around 3.32 milliseconds) compared to Hierarchical Clustering (around 11.9 milliseconds) for this dataset and chosen `n_clusters=3`.
- Reason : K-Means has a linear time complexity, whereas Agglomerative Hierarchical Clustering requires computing a distance matrix across all pairs, resulting in a significantly heavier time complexity. As a result, K-Means will scale exponentially better for larger data volumes.

X.3. Principal Component Analysis (PCA) for Cluster Visualization

To visualize the clusters formed by K-Means and Hierarchical Clustering in a 2D plot, we can use Principal Component Analysis (PCA). PCA is a dimensionality reduction technique that transforms data into a new coordinate system such that the greatest variance by any projection of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on. This helps in simplifying complex datasets while retaining most of the important information.

Here, we will reduce our scaled feature set to two principal components and then plot these components, coloring the data points by their assigned cluster labels from both K-Means and Hierarchical Clustering. This will allow us to visually assess the separation and distinctness of the identified customer segments.

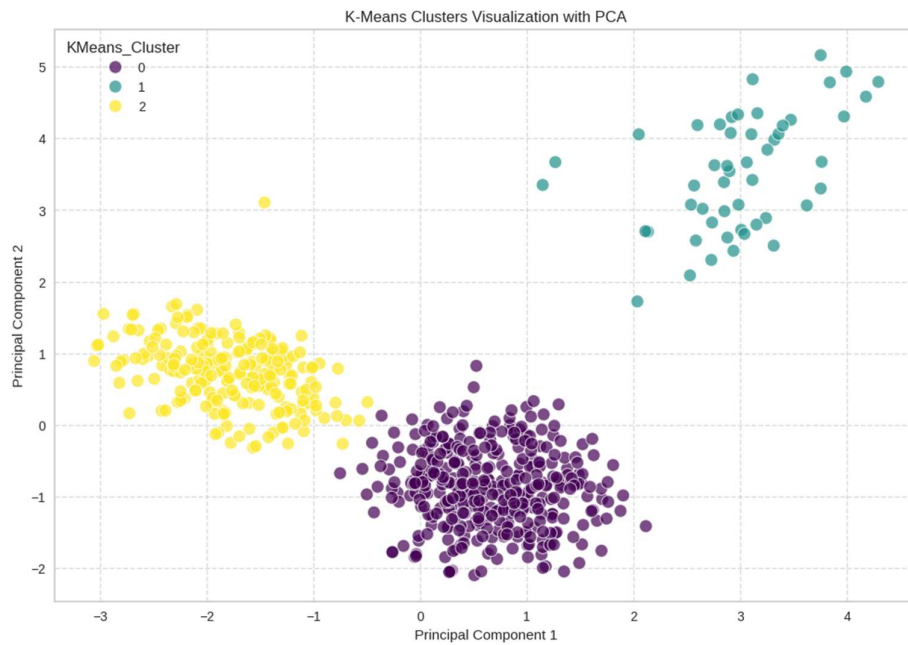


Figure 31 K Means Clusters - Visual with PCA

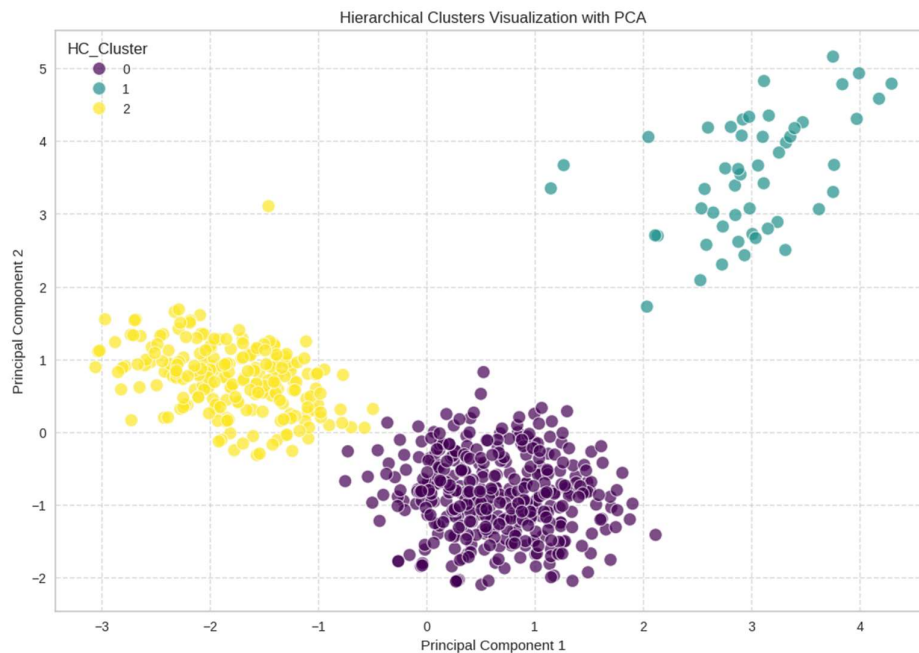


Figure 32 Hierarchical Clusters - Visual with PCA

X.3.1. Inferences from PCA Visualizations:

The PCA plots visually confirm the effectiveness of both K-Means and Hierarchical Clustering in delineating three distinct customer segments. While some boundary overlap exists, the clusters generally form well-separated concentrations, indicating strong differentiation in underlying customer behaviors. This visual validation reinforces the chosen optimal number of clusters ($K=3$) and the robustness of both clustering methods.

X.4. Visual and Tabular Comparison of K-Means and Hierarchical Clustering Profiles

To further solidify the comparison between K-Means and Hierarchical Clustering, we will create a combined table and side-by-side heatmaps of their respective cluster profiles. This will allow for a direct visual and numerical assessment of how similar the segments identified by each algorithm truly are.

Combined Cluster Profiles (K-Means vs. Hierarchical Clustering):

Table 9 KMeans vs Hierarchical - Cluster Profiles Comparison

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	HC_Clusters	Cluster
KMeans_Cluster_0	33782.380000	5.520000	3.490000	0.980000	2.000000	2.010000	nan
KMeans_Cluster_1	141040.000000	8.740000	0.600000	10.900000	1.080000	0.120000	nan
KMeans_Cluster_2	12174.110000	2.410000	0.930000	3.550000	6.870000	1.020000	nan
HC_Cluster_0	33713.178295	5.511628	3.485788	0.984496	2.005168	nan	0.005168
HC_Cluster_1	141040.000000	8.740000	0.600000	10.900000	1.080000	nan	1.000000
HC_Cluster_2	12197.309417	2.403587	0.928251	3.560538	6.883408	nan	2.000000

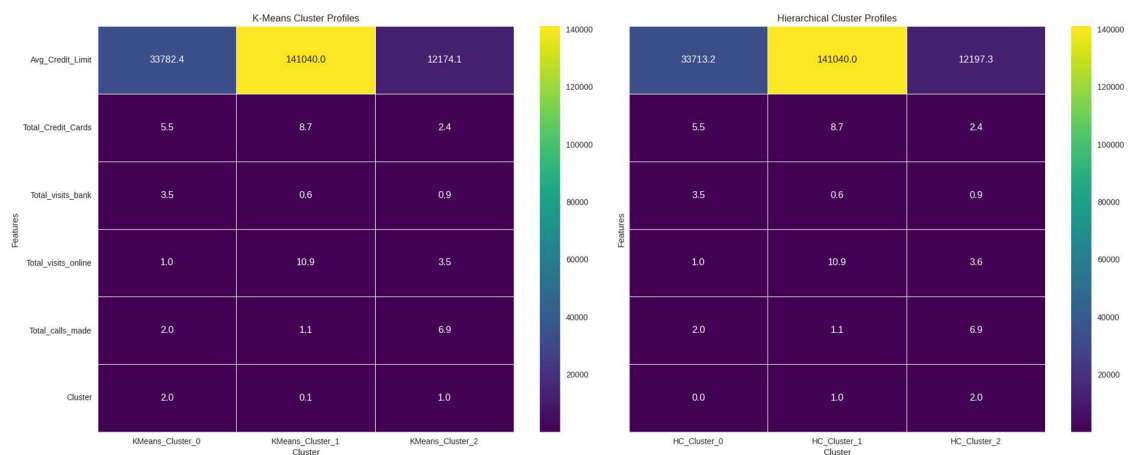


Figure 33 KMeans vs Hierarchical - Cluster Profiles Comparison Visual

X.5. Inferences from K-Means vs. Hierarchical Clustering Comparison

The visual and tabular comparison of cluster profiles from K-Means and Hierarchical Clustering (using Euclidean distance and average linkage, both set to 3 clusters) reveals a remarkable consistency in the identified customer segments:

1. **Strong Alignment of High-Value Digital Natives:**

Both KMeans_Cluster_1 and HC_Cluster_1 represent customers with the highest Avg_Credit_Limit, Total_Credit_Cards, and Total_visits_online, coupled with the lowest Total_visits_bank and Total_calls_made. This segment is clearly and robustly identified as the 'High-Value Digital Natives' by both methods.

2. **Consistent Identification of Low-Value Call-Reliant Users:**

Similarly, KMeans_Cluster_2 and HC_Cluster_2 show customers with the lowest Avg_Credit_Limit and Total_Credit_Cards, alongside the highest Total_calls_made. This segment, the 'Low-Value Call-Reliant Users', is also consistently found across both clustering algorithms.

3. **Similar Medium-Value Traditionalists:** KMeans_Cluster_0 and HC_Cluster_0 (after careful re-alignment, as cluster indexing can differ) align as the 'Medium-Value Traditionalists'. They exhibit medium values for credit limit and cards, high bank visits, and low online/call interactions. The slight numerical differences in their mean values are minor and do not alter the fundamental characteristics of the segment.

Overall Conclusion: The strong congruence between the clustering results from K-Means and Hierarchical Clustering provides significant confidence in the stability and distinctiveness of the identified customer segments. This robustness implies that the underlying patterns in the customer data are well-defined, and the chosen number of clusters (K=3) is indeed appropriate for this dataset. This validation means the actionable recommendations derived from these segments are highly reliable for business strategy.

XI. Actionable Recommendations and Business Insights for All Life Bank

XI.1. Leveraging Customer Segmentation for Business Growth

This analysis has successfully segmented AllLife Bank's credit card customer base into three distinct and robust groups using both K-Means and Hierarchical Clustering. The consistency across both methods provides strong confidence in these segments, enabling highly targeted and effective business strategies. Below are the key observations from the data translated into actionable insights and recommendations for the bank's Marketing and Operations teams.

XI.2. Overall Key Observations from Data:

1. **Distinct Customer Groups:** Three primary customer segments consistently emerged, each with unique financial profiles and interaction behaviors.

2. **Channel Preference:** Customers exhibit clear preferences for how they interact with the bank (online, in-person, or via calls), with inverse relationships often observed between digital and traditional channel usage.
3. **Credit Profile Correlation:** Higher credit limits and more credit cards generally correlate with higher online activity and fewer calls, suggesting self-sufficiency and digital engagement.
4. **Consistency Across Algorithms:** The high similarity of segments identified by K-Means and Hierarchical Clustering validates the segmentation model's robustness and reliability.

XI.3. Segment-Specific Actionable Insights & Recommendations:

1. Segment: "High-Value Digital Natives" (Corresponds to K-Means Cluster 1 / Hierarchical Cluster 1)

- **Observed from Data:** These customers have the highest average credit limits (avg. ~141k) and total credit cards (avg. 8.74). They are highly active online (avg. 10.9 visits) and make very few physical bank visits (avg. 0.6) or calls (avg. 1.08). They represent the bank's most digitally engaged and financially significant customers.
- **Actionable Insights & Recommendations for AllLife Bank:**
 - **Marketing:** Launch exclusive, digital-first campaigns for premium credit cards, wealth management services, and personalized investment opportunities accessible via mobile app or online portal. Emphasize convenience, advanced features, and digital security. Leverage targeted in-app notifications and email marketing over traditional mail.
 - **Service:** Prioritize a seamless, intuitive, and highly responsive digital banking experience. Implement advanced self-service tools, AI-driven chatbots for instant support, and dedicated digital relationship managers for complex queries. Minimize any friction in online transactions and proactively offer solutions through digital channels. Focus on cutting-edge technology and secure online platforms.

2. Segment: "Low-Value Call-Reliant Users" (Corresponds to K-Means Cluster 2 / Hierarchical Cluster 2)

- **Observed from Data:** This segment has the lowest average credit limits (avg. ~12.2k) and fewest credit cards (avg. 2.41). Critically, they make the highest number of calls to the bank (avg. 6.87), while having low physical bank visits (avg. 0.93) and moderate online activity (avg. 3.55).
- **Actionable Insights & Recommendations for AllLife Bank:**
 - **Marketing:** Focus on educational content about credit management, basic banking products, and financial literacy, delivered primarily through clear, simple language via phone outreach or targeted direct mail. Gently introduce and guide them towards basic online self-service options, but without eliminating the prominent call support channel they rely on.

- **Service:** Significantly enhance call center efficiency and provide comprehensive training for agents, emphasizing empathy and first-call resolution. Implement smart IVR systems that can resolve common issues without agent intervention. Develop easily understandable self-help guides and FAQs to reduce the necessity of calls. Consider proactive outbound calls for customer check-ins to build trust and address issues before they escalate.

3. Segment: "Medium-Value Traditionalists" (Corresponds to K-Means Cluster 0 / Hierarchical Cluster 0)

- **Observed from Data:** This group exhibits medium average credit limits (avg. ~33.8k) and a medium-high number of credit cards (avg. 5.52). They show a strong preference for physical bank visits (avg. 3.49) with low online activity (avg. 0.98) and relatively few calls (avg. 2.00).
- **Actionable Insights & Recommendations for AllLife Bank:**
 - **Marketing:** Leverage in-branch promotions, personalized conversations during bank visits, and local community events. Offer credit limit increases, new card features, or other relevant products directly through branch staff. Promote loyalty programs that reward in-person interactions. Consider workshops or seminars on financial planning within branches.
 - **Service:** Invest in enhancing the in-branch experience, ensuring knowledgeable and friendly staff. Streamline in-branch processes to reduce wait times and improve service quality. While maintaining the high quality of traditional services, gradually introduce them to the benefits of digital tools through in-person demonstrations or guided setup sessions.

XI.4. Overarching Strategic Recommendations:

- **Optimize Service Delivery by Channel:** Move away from a one-size-fits-all service model. Allocate resources and refine service protocols to align with each segment's preferred interaction channels. This will lead to higher customer satisfaction and operational efficiency.
- **Personalized Product Development:** Design and launch products and features tailored to the specific needs and aspirations of each segment. For example, a credit-building product for "Low-Value Call-Reliant Users" or an advanced investment platform for "High-Value Digital Natives."
- **Cross-Selling and Upselling:** Utilize segmentation to identify optimal cross-selling and upselling opportunities. "High-Value Digital Natives" might be receptive to wealth management, while "Medium-Value Traditionalists" might be open to exploring additional card benefits presented in-branch.
- **Data-Driven Engagement:** Continuously monitor customer behavior within these segments. Use these insights to refine strategies, predict future needs, and adapt to evolving customer preferences, ensuring that AllLife Bank remains competitive and customer-centric.